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HER2-driven mammary tumorigenesis enhances bioenergetics despite reductions in mitochondrial content

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eLife Assessment

This **valuable** study aims to determine mechanisms underlying breast cancer initiation and tumour progression. The manuscript includes a **solid** set of transcriptomic and proteomic datasets from tumour samples and examines mitochondrial function within the tumours. While the underlying mechanisms linking expression changes to functional effects remain speculative. This paper provides a resource for researchers working on breast cancer and/or HER2-driven bioenergetics changes.

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Abstract

It is now recognized that mitochondria play a crucial role in tumorigenesis, however, it has become clear that tumor metabolism varies significantly between cancer types. The failure of recent clinical trials aimed at directly targeting tumor respiration through oxidative phosphorylation inhibitors underscores the critical need for further studies providing an in-depth evaluation of mitochondrial bioenergetics. Accordingly, we comprehensively assessed the bulk tumor and mitochondrial metabolic phenotype in murine HER2-driven mammary cancer tumors and benign mammary tissue. Transcriptomic and proteomic profiling revealed a broad downregulation of mitochondrial genes/proteins in tumors, including OXPHOS subunits comprising Complexes I-IV. Despite reductions in tumor mitochondrial proteins, mitochondrial respiration was several-fold higher compared to benign mammary tissue, which persisted regardless of normalization method (wet weight, total protein content and when corrected for mitochondrial content). This upregulated respiratory capacity could not be explained by OXPHOS uncoupling, suggesting HER2 signaling regulates intrinsic mitochondrial bioenergetics. In further support, lapatinib, an EGFR/HER2 tyrosine kinase inhibitor, attenuated mitochondrial respiration in NF639 murine mammary tumor epithelial cells. Together, this data highlights that the typical correlation between mitochondrial content and respiratory capacity may not apply to all tumor types and implicates HER2-linked activation of mitochondrial respiration supporting tumorigenesis in this model.

Introduction

The field of cancer metabolism has rapidly evolved in the last decade. A specific focus of many researchers has been on understanding the role of mitochondria in tumorigenesis with the goal to potentially target aspects of mitochondrial metabolism to reduce proliferation^{1,2}. It was initially hypothesized that a universal characteristic of cancer cells was dysfunctional oxidative phosphorylation (OXPHOS), as proposed by Otto Warburg in the 1920s^{3,4}. This theory spurred interest in studying mitochondria in various cancers, highlighting the diverse mitochondrial phenotypes in this context. While instrumental for OXPHOS, mitochondria are now recognized for their multifaceted roles in regulating cellular homeostasis, including supporting macromolecule biosynthesis⁵, maintaining redox balance⁶ and generating metabolites that regulate gene transcription⁷. Given their pleiotropic effects within the cell, there is growing recognition that mitochondria play an essential role in cancer initiation, progression, and metastasis¹, and an appreciation that interrogating mitochondrial bioenergetics in diverse cancers may identify novel therapeutic strategies.

Reductions in the content/activity of mitochondrial enzymes within the tricarboxylic acid (TCA) cycle and electron transport chain (ETC) often correlate with impaired mitochondrial respiratory capacity and coincides with activating transcriptional programs driving tumorigenesis^{8–10}. Indeed, inhibition of succinate dehydrogenase (Complex II) in osteosarcoma cells⁸, and renal cell carcinoma⁹ enhances cancer growth despite OXPHOS inhibition. Furthermore, in colorectal cancer, age-related Complex I deficiency coincides with reduced respiratory capacity but enhanced serine synthesis and subsequent growth rates¹⁰. Despite this evidence suggesting that disruptions in mitochondrial bioenergetics contribute to tumor progression, several elegant studies using genetic and pharmacological approaches have solidified that a functional ETC is required for cancer cell growth^{6,11,12}. Furthermore, the use of whole-genome sequencing specifically interrogating mtDNA phenotypes across several human cancers has revealed that loss of function mutations in mtDNA genes are often under negative selection¹³, suggesting the maintenance of mitochondrial function may support tumorigenesis in some cancers and impairments in mitochondrial metabolism are not a ubiquitous finding. However, mitochondrial content represents one dimension of bioenergetic regulation and does not alone determine respiratory capacity. Therefore, a comprehensive assessment of mitochondrial function, particularly within the context of the mitochondrial proteome, is warranted to provide deeper insights into the relationship between mitochondrial content and bioenergetics in different cancers.

Within this framework, there has been considerable interest in investigating mitochondrial bioenergetics and dynamics in breast cancer subtypes due to the limited therapeutic options for triple-negative breast cancer (TNBC) and therapeutic resistance in breast cancers driven by estrogen (ER), progesterone (PR) and human epidermal growth factor (HER2) receptor amplification^{14–16}. Indeed, increases in mitochondrial respiration or “OXPHOS reliance” is associated with poor outcomes in treatment naïve TNBC¹⁷, metastatic ER+ tumors¹⁸, and HER2-driven breast cancers¹⁹. While inhibiting OXPHOS in this context^{17,18} (i.e. with Complex I inhibitor IACS-010759) has shown some pre-clinical success, systemic toxicity has prevented the translation of this strategy to the clinic²⁰. Furthermore, it is unclear whether increases in mitochondrial respiration arise from increases in mitochondrial content or intrinsic enhancements in mitochondrial function (e.g. coupling efficiency, substrate delivery/sensitivity, oncogenic signaling events). This information is critical to develop successful mitochondrial-targeted therapies with a wider therapeutic window. Thus, in the current study, we comprehensively assessed the bulk tumor and mitochondrial phenotype between murine HER2-driven mammary cancer tumors and benign mammary tissue in the context of the underlying transcriptome and proteome. Respiratory capacity supported by both pyruvate and lipids was enhanced in HER2-driven tumors despite lower mitochondrial content compared to benign mammary tissue. Mitochondrial bioenergetics were suppressed by the dual tyrosine kinase inhibitor lapatinib, suggesting events linked to HER2-mediated oncogenic signaling may drive increases in OXPHOS during tumorigenesis.

Results

Mammary tumors from MMTV/neu^{ndl}-YD5 mice display canonical HER2 pathway activation

Cell lines are commonly used to study isolated mechanisms driving tumorigenesis, however, they neglect the variety of cell types in the tissue of origin and the influence of the tumor microenvironment, which is a key regulator of cancer growth^{21,22}. To this end, we have utilized tumor and benign mammary tissues from MMTV/neu^{ndl}-YD5 transgenic mice²³ to investigate HER2-driven tumor biology (Fig.1A [↗](#)). Signaling events linked to HER2 activation rely on the autophosphorylation of the tyrosine kinase domain^{23–25} and MMTV/neu^{ndl}-YD5 mice possess an add-back mutation at the Y1226/7 (YD) site, activating downstream signaling events linked to SHC/GRB2 with similar transforming potential as the wild-type protein^{25,26} (Fig.1B [↗](#)). While the proximal signaling events linked to HER2 activation in this model have been described²⁵, to validate our bulk tissue-based model, we assessed the mRNA (transcriptomics) and protein (quantitative label-free proteomics and western blotting) expression of canonical HER2 pathway members (RAS/MAPK and PI3K/AKT/mTOR pathways) that are well-characterized in various HER2-overexpressing mammary cancer cell lines (Fig.1B [↗](#),^{27–29}). *Her2* transcript levels were increased ~175-fold (Fig.1C [↗](#)) and transcripts belonging to the Ras/MAPK (*Hras*, *Mapk1*, *Mapk3*) and PI3K/AKT/mTOR (*Pik3cg*, *Pik3c2a*, *Pik3c2b*, *Pikrcg*, *Pik3c3*, *Akt3*) pathways were also upregulated in the tumors compared to the benign mammary tissue (Fig.1D [↗](#)). At the protein level, HER2 was readily detectable in mammary tumors by quantitative label-free proteomics but undetectable in paired benign tissues (Fig.1E [↗](#)) suggesting enhanced HER2 transcription also manifests as an increase in receptor content in the tumors. Downstream of HER2 activation, the total protein content of p38 MAPK, ERK1/2 and eEF2 were significantly increased in tumors compared to benign tissue detected by western blot (Fig.1F,G [↗](#)). When normalized to total protein, p38 MAPK and eEF2 phosphorylation were not higher in tumors due to proportional increases in total protein abundance, while in contrast, phosphorylated ERK1/2 and mTOR were significantly upregulated suggesting both Ras/MAPK and the PI3K/AKT/mTOR pathways were activated in tumors (Fig.1H [↗](#), S-Fig.1 [↗](#)). Overall, these data demonstrate that mammary tumors display classical hallmarks of HER2-driven cancer cell signaling, allowing for the subsequent investigation of HER2-driven tumor biology and metabolism in this model.

Assessment of the HER2-driven mammary tumor transcriptome

To understand the biology and metabolism of HER2-driven mammary tumors, we assessed their transcriptomes using RNA sequencing. Principal component analysis (PCA) using the entire gene expression profile showed distinct separation between groups (Fig.2A [↗](#)). Visualizing all differentially expressed genes (DEGs) using a volcano plot (Fig.2B [↗](#)) and the top 3000 DEGs using a heatmap (Fig.2C [↗](#)) revealed several differentially upregulated and downregulated genes between groups where genes involved in MAPK- and AKT-linked signaling, ECM remodelling and cellular stress responses (*Etv1*, *Dusp6*, *Papln*, *Ier3*) were among the most upregulated genes in the tumors and genes involved in lipid metabolism (*Dgat2*, *Lipe*, *Cd36*) were among the most downregulated. Enrichment analysis of the top 3000 upregulated DEGs (Fig.2D [↗](#)) identified pathways characteristic of HER2-driven signaling (e.g. tyrosine kinase signaling, GTPase signaling) and gene ontology (GO) analysis identified processes indicative of active transcription, translation, and protein turnover characteristic of a rapidly proliferating tissue (Fig.2E [↗](#)). Interestingly, enrichment analysis of the top 3000 downregulated genes in the tumors consistently identified that mitochondrial metabolism, and in particular components of the oxidative phosphorylation (OXPHOS) system, were downregulated in tumors compared to benign mammary tissue (Fig.2F [↗](#),G).

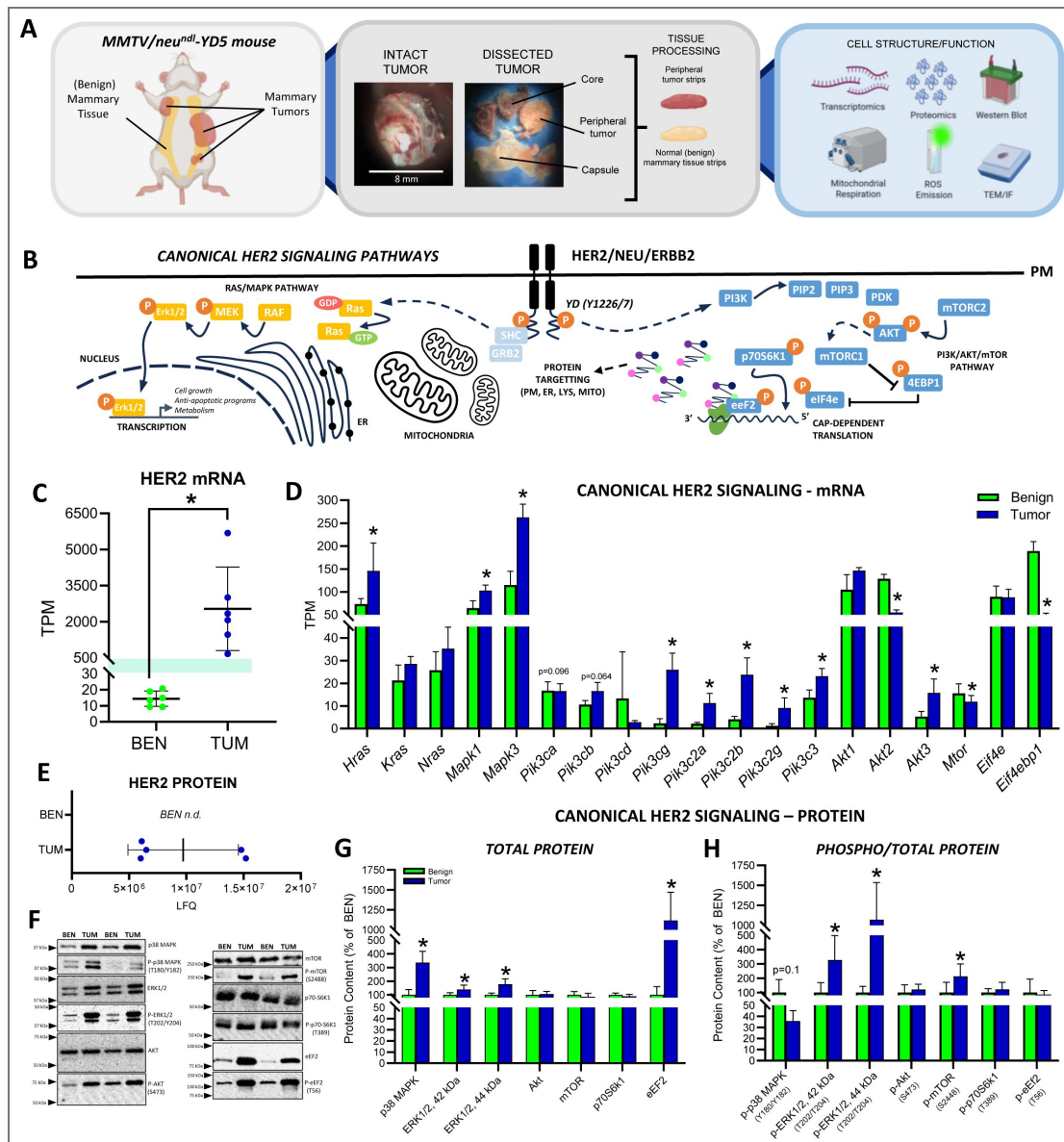


Figure 1. Experimental design and canonical HER2 signaling.

A) Schematic of mammary tumors from the mammary fat pad of female MMTV-neu^{ndL}-YD5 mice and normal (benign) mammary tissue, representative images of tumor morphology (tumor capsule, peripheral tumor tissue and necrotic core), tissue processing workflow for transcriptomics, proteomics, western blotting, mitochondrial respiration, ROS emission, and histology (transmission electron microscopy (TEM) and immunofluorescence (IF)). **B)** Schematic of classical signaling pathways (RAS/MAPK and PI3K/AKT/mTOR) downstream of HER2 receptor stimulation in the YD5 model. **C)** *Her2* gene expression (transcripts per million, TPM), **D)** Gene expression for canonical HER2 signaling. **E)** Absolute abundance of HER2 protein detected by quantitative label-free proteomics (n.d. = HER2 not detected in benign). **F)** Representative western blots of proteins involved in canonical HER2 signaling. **G,H)** Quantified western blots (% of benign) for **G)** total protein content and **H)** phosphorylated targets normalized to the total protein content of each target. P-values in panels **D** and **E** listed as adjusted p values from transcriptomic (DEG) data (p-adj. <0.1 considered statistically significant). In panels **G,H** data is presented as means $\pm$ SD and statistical significance was determined using an unpaired, two tailed Student's t-test (*= p <0.05, n =5-6 per group/protein target). **Abbrev.** AKT - protein kinase B; eIF2 - eukaryotic elongation factor 2; eIF4E - eukaryotic translation initiation factor 4E; ERK - extracellular signal-regulated kinase; GRB2 - growth factor receptor-bound protein-2; MEK - mitogen-activated protein kinase kinase; mTORC1/2 - mechanistic target of rapamycin complex 1 and 2; PDK - 3-phosphoinositide-dependent protein kinase-1; PI3K - phosphoinositide 3-kinase; PIP2/3 - phosphatidylinositol 4,5 bisphosphate, PIP3 - phosphatidylinositol 3,4,5-triphosphate; p70S6K1 - ribosomal protein S6 kinase beta-1; RAF - rapidly accelerated fibrosarcoma; RAS - rate sarcoma virus; SHC - Src homolog and collagen homolog; 4EBP1 - eukaryotic translation initiation factor 4E-binding protein 1.

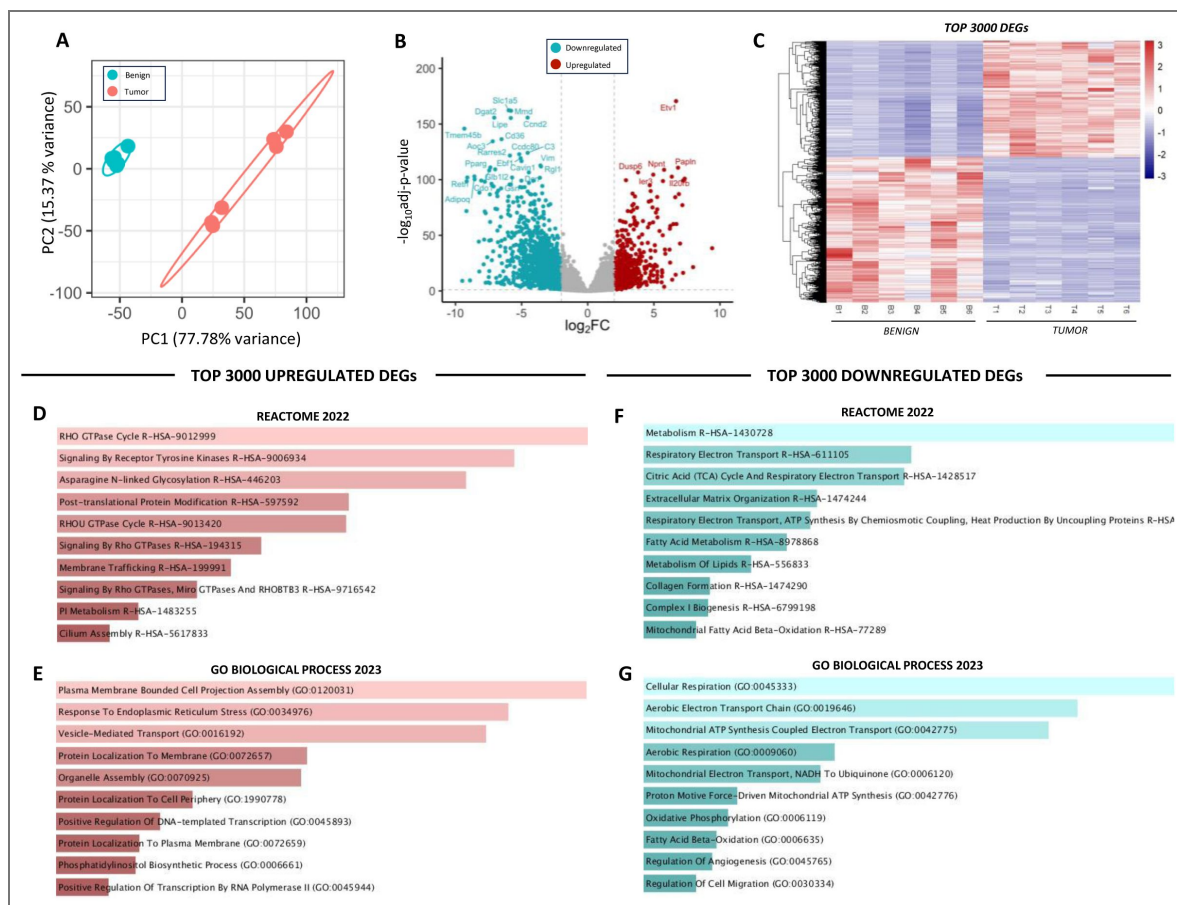


Figure 2. Assessment of HER2-driven mammary tumor and benign mammary tissue transcriptomes.

A) PCA of benign mammary tissue and mammary tumors, **B**) Volcano plot showing differentially expressed genes (DEGs) identified between mammary tissue and mammary tumors. **C**) Heatmap of top 3000 DEGs, **D**) Top 10 upregulated pathways (Reactome 2022) and **E**) gene ontologies (GO Biological Process 2023) in tumors compared to benign mammary tissue, **F**) Top 10 downregulated pathways (Reactome 2022) and **G**) gene ontologies (GO Biological Process 2023) in tumors compared to benign mammary tissue. Panels **D,E** assess the top 3000 upregulated DEGs ($\log_2\text{fc}>1$), panels **F,G** assess the top 3000 downregulated DEGs ($\log_2\text{fc}<1$). An adjusted p-value of <0.1 was considered statistically significant across all analyses. The Enrichr open access analysis web tool was used to generate panels **D-G** which were sorted by p-value.

Assessment of HER2-driven mammary tumor proteome

We further validated our bulk-tissue transcriptomic findings at the protein level using label-free quantitative proteomics and detected 2589 proteins in the benign mammary tissue and 2448 proteins in the HER2-driven tumors; 1848 of which were found in both groups (Fig.3A). This initial analysis revealed 741 and 600 uniquely expressed (i.e. only detected in one tissue) proteins in benign mammary tissue and tumors respectively (S-Fig.2A). Interestingly, OXPHOS and aerobic respiration were identified among the top pathways in benign tissue, but not in tumors (S-Fig.2B-G), further supporting the notion of downregulated mitochondrial content within tumors. Reflecting the PCA for the transcriptomic data, analysis of all overlapping proteins between benign and tumor samples revealed distinct separation (Fig.3B). Of the 1848 proteins found in both the tumors and benign tissue, 420 were significantly different (p-adj. <0.1, Fig. 3A,C,D). However, the majority of the differentially expressed proteins were downregulated (n=411) in tumors compared to benign mammary tissue. Upregulated proteins (n=9) included those involved in canonical HER2 signaling, protein synthesis and processing, and glycolytic metabolism (S-Table 1). Supporting transcriptomic data, which indicated a reduction in mitochondrial content and OXPHOS, enrichment analysis revealed that 8 of the top 10 downregulated pathways (Fig. 3E) and all top 10 downregulated GO (Fig. 3F) were related to mitochondrial biogenesis and/or metabolism, and OXPHOS was identified in the top 10 downregulated KEGG pathways (Fig.3G). Together, these proteomic data support our transcriptomic analysis suggesting HER2-driven tumors are characterized by lower mitochondrial content than benign mammary tissue.

Integrated assessment of the HER2-driven tumor transcriptome and proteome

The concordant nature of our transcriptomic and proteomic datasets was highlighted when correlating the log2FC between overlapping gene and protein targets (Fig.4A). Sorting these data by mitochondrial proteins highlighted that the majority of overlapping mitochondrial proteins detected in both datasets were downregulated (Fig.4B). Thus, we further analyzed the 144 overlapping downregulated targets between these analyses (Fig.4C). KEGG enrichment analysis of the downregulated targets identified OXPHOS among the top 10 pathways, as well as carbon metabolism, BCAA degradation and fatty acid metabolism which encompass numerous mitochondrial targets (Fig.4D). The top 10 common downregulated pathways (Fig.4E) and GO (Fig.4F) all pertained to mitochondrial substrate metabolism/catabolism and OXPHOS. Together, transcriptomic and proteomic analyses strongly support that HER2-driven tumors are characterized by downregulated mitochondrial content.

Assessment of the mitochondrial transcriptome and proteome

Our initial transcriptomic and proteomic analyses encompassed all differentially expressed genes/proteins between tissues (including mitochondrial targets) and clearly identified reductions in mitochondrial content in the tumors. This emphasized the robust nature of mitochondrial adaptations in this model, considering mitochondrial proteins (identified using the mouse MitoCarta 3.0 database) represented a small fraction of total cellular protein in both tumors and benign mammary tissue (S-Fig.3A). By filtering our proteomic dataset using MitoCarta, we identified 125 mitochondrial proteins that were commonly expressed between the tumors and benign mammary tissue (Fig.5A). When visualizing the 125 proteins with a heatmap, the majority of the mitochondrial proteome was downregulated in the tumors compared to the benign mammary tissue (Fig.5B). The summed absolute abundance of all detected mitochondrial proteins in each group was also lower in tumors (Fig.5C), again supporting our transcriptomic and proteomic analyses suggesting a reduction in mitochondrial content exists in the tumors.

While mitochondrial biogenesis is a coordinated process simultaneously upregulating several mitochondrial proteins, the total mitochondrial transcriptome/proteome does not necessarily change proportionally with OXPHOS complex and/or TCA cycle/dehydrogenase content, especially

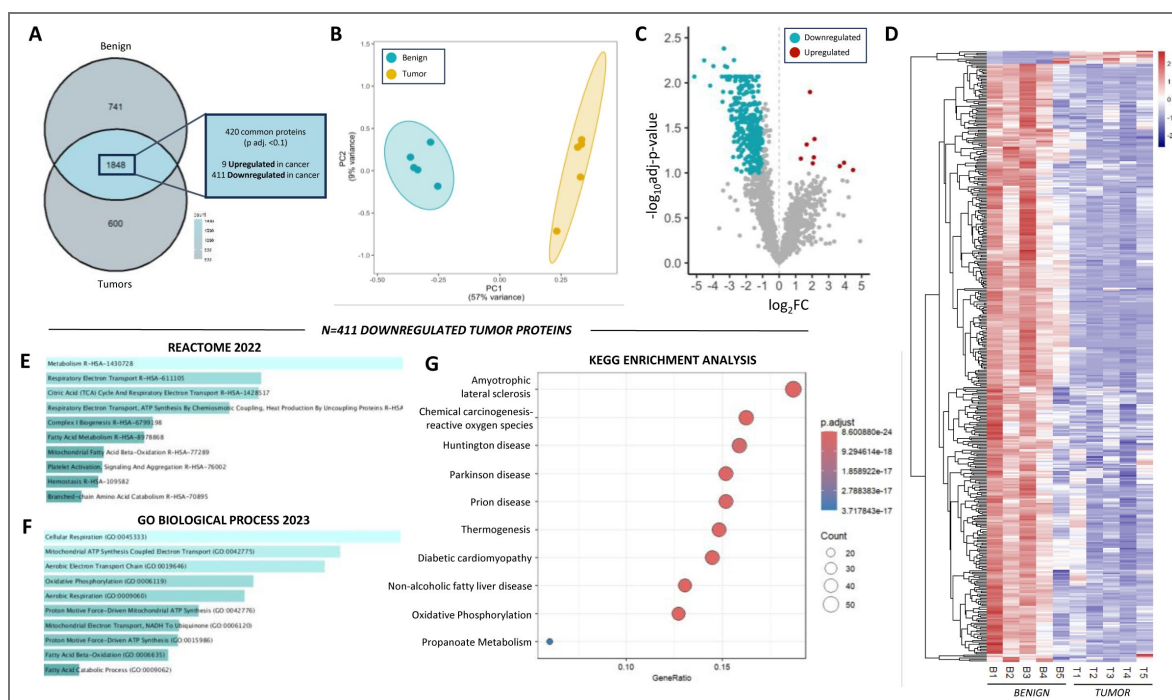


Figure 3. Assessment of HER2-driven mammary tumor and benign mammary tissue proteomes.

A) Venn diagram showing all detected proteins. Of the 1848 overlapping targets, 420 were significantly different based on a p-adj. <0.1 and 411 of those proteins were downregulated in tumors. **B)** PCA of benign mammary samples and mammary tumors, **C)** Volcano plot showing proteins identified between mammary tumors and benign mammary tissue, those in blue (downregulated) or red (upregulated) were significantly different (n=420 proteins, p-adj. <0.1). **D)** Heatmap of 420 differentially expressed proteins in benign and tumor samples. **E)** Top 10 downregulated pathways (Reactome 2022) and **F)** gene ontologies (GO Biological Process, 2023), **G)** KEGG analysis of downregulated proteins. The Enrichr open access analysis web tool was used to generate panels **E,F** which were sorted by p-value.

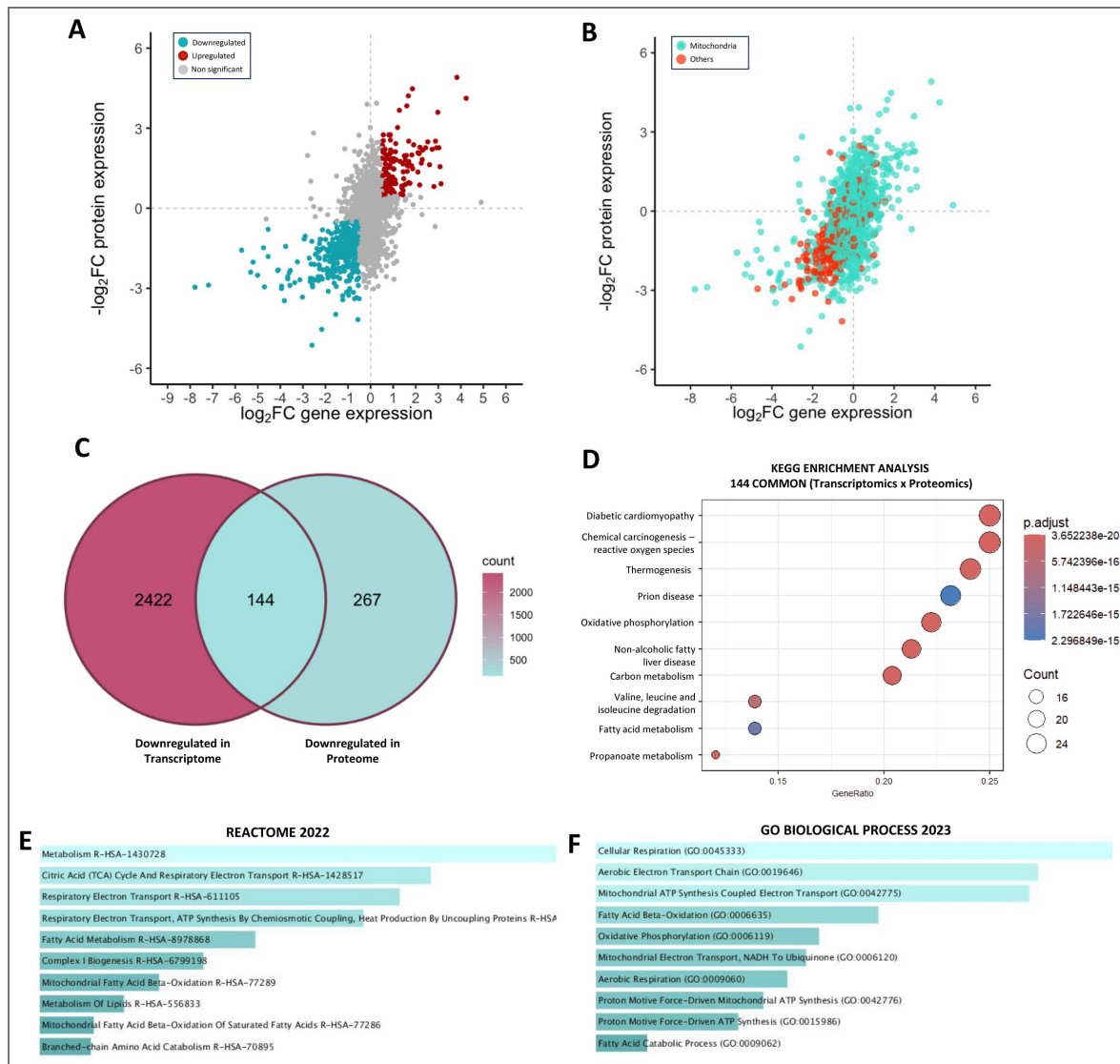


Figure 4. Overlapping genes and proteins between HER2-driven mammary tumor transcriptome and proteome.

Correlation of log2FC for overlapping targets between the HER2-driven tumor and benign mammary tissue transcriptomes and proteomes grouped by up/downregulation **A**, and **B**) protein localization (mitochondrial vs other). **C**) Venn diagram showing 144 overlapping downregulated targets between the assessed HER2-driven tumor and benign mammary tissue transcriptomes and proteomes. **D**) KEGG analysis for common downregulated targets. Top 10 downregulated **E**) pathways (Reactome 2022) and **F**) gene ontologies (GO Biological Process 2023). An adjusted p value of <0.1 was utilized for all analyses. The Enrichr open access analysis web tool was used to generate panels **E**, **F** which were sorted by p-value.

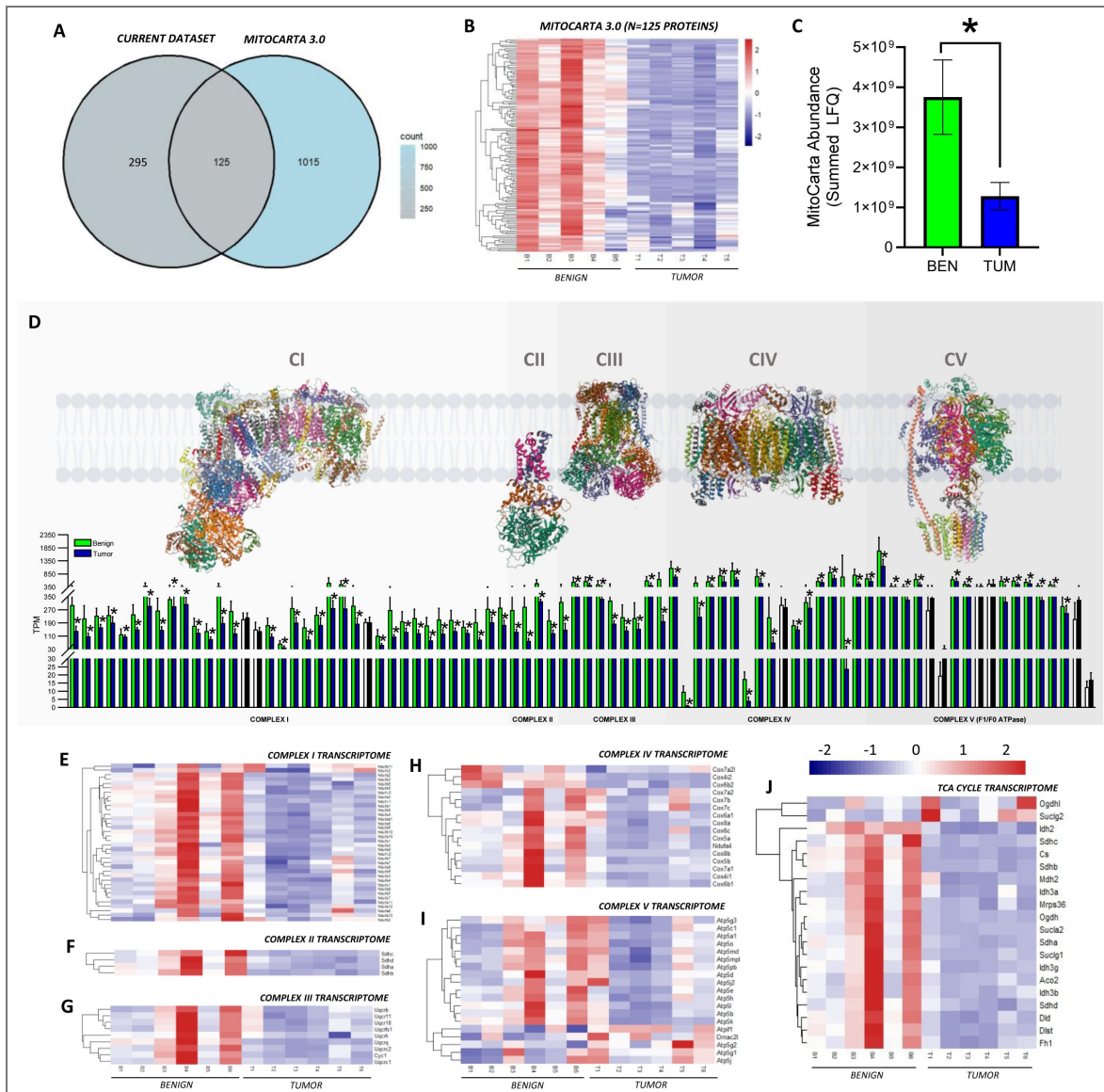


Figure 5. Assessment of the mitochondrial transcriptome and proteome.

A) Venn diagram showing mitochondrial proteins detected in the current dataset when cross-referenced with the Mouse MitoCarta 3.0 database. **B)** Heatmap of 125 mitochondrial proteins identified in benign mammary tissue and HER2-driven tumors (n=5/group). **C)** Absolute abundance of all mitochondrial proteins detected in benign and tumor tissues. **D)** Gene expression (detected by transcriptomics; transcripts per million) of all detected OXPHOS subunits separated per respiratory complex, black and white bars represent subunits with no significant differences between groups. **E-J)** Heatmaps of **E)** Complex I, **F)** Complex II, **G)** Complex III, **H)** Complex IV, **I)** Complex V (ATP synthase) and **J)** TCA cycle transcriptomes. All proteomic analyses (**panels A-C**) were conducted on n=5/group. All transcriptomic analyses (**panels D-J**) were conducted on n=6/group. **Panels B,C** were analyzed using an unpaired two tailed Student's t-test (*p<0.05). In **Panel D**, * and coloured (blue and green bars) represent significant differences between benign and tumor samples based on an adjusted p value of <0.1. OXPHOS complex schematics in panel **D** were obtained from the open access RCSB Protein Data Bank (Complex I = 1W1V, Complex II = 2A06, Complex III = 3SFD, Complex IV = 6NMP, Complex V = 5FIL). Bars in panel **D** are listed in the following order: Complex I: Ndufa1, Ndufa10, Ndufa11, Ndufa12, Ndufa13, Ndufa2, Ndufa3, Ndufa5, Ndufa6, Ndufa7, Ndufa8, Ndufa9, Ndufab1, Ndufb10, Ndufb11, Ndufb2, Ndufb3, Ndufb4, Ndufb5, Ndufb6, Ndufb7, Ndufb8, Ndufb9, Ndufc1, Ndufc2, Ndufs1, Ndufs2, Ndufs3, Ndufs4, Ndufs5, Ndufs6, Ndufs7, Ndufs8, Ndufv1, Ndufv2, Ndufv3; Complex II: Sdha, Sdhb, Sdhc, Sdhd; Complex III: Cyc1, Uqcrl10, Uqcrl11, Uqcrb, Uqcrcl1, Uqcrcl2, Uqcrf1, Uqcrh, Uqcrq; Complex IV: Cox4i1, Cox4i2, Cox5a, Cox5b, Cox6a1, Cox6b1, Cox6b2, Cox6c, Cox7a1, Cox7a2, Cox7a2l, Cox7b, Cox7c, Cox8a, Cox8b, Ndufa4; Complex V: Atp5a1, Atp5b, Atp5c1, Atp5d, Atp5e, Atp5g1, Atp5g2, Atp5g3, Atp5h, Atp5j, Atp5j2, Atp5k, Atp5l, Atp5md, Atp5mpl, Atp5o, Atp5pb, Atpf1, Dmac2l.

in cancerous tissues. Indeed, reductions in the content of certain respiratory complexes seem to be cancer specific, which can have functional repercussions on mitochondrial respiration rates and/or efficiency^{10,31}. Within this context, we assessed the gene expression profile of all OXPHOS complex subunits between groups, which revealed relatively ubiquitous reductions in transcript count (transcripts per million) across subunits of Complexes I-IV in the tumors (Fig. 5D-H). Interestingly, while the majority of Complex V subunits were also downregulated, this complex had the highest proportion of subunits displaying no differences between groups (Fig. 5D,I). The TCA cycle transcriptome also reflected a general downregulation of transcript abundance, apart from alpha-ketoglutarate dehydrogenase (*Ogdh*) and succinyl-CoA synthetase GDP forming subunit- β (*Suc1g2*), which had a higher transcript abundance in some tumor samples (Fig. 5J). While we focused on transcriptomic data for this analysis due to lower detection of certain OXPHOS subunits using label-free quantitative proteomics (S-Fig. 4A-E), the summed abundance of all OXPHOS subunits detected per complex expressed relative to the underlying mitochondrial proteome (S-Fig. 4F), and relative to the total proteome (S-Fig. 4G) displayed a similar pattern. This interpretation was further validated by western blot, where tumor CII and CIV content were lower than benign mammary tissue (S-Fig. 4H). Overall, these data further suggest that the mitochondrial proteome and OXPHOS Complexes I-IV are downregulated in HER2-driven tumors.

While these findings strongly support reduced mitochondrial gene expression in tumors, differences could theoretically result from shifts in cell-type during tumorigenesis. To address this, we first assessed OXPHOS expression in a publicly available single-cell RNA sequencing (scRNA-seq) dataset from human breast tumors³¹. In the scRNA-seq dataset, tumor epithelial cells expressed the highest proportion of CI-CIV subunits when grouped by cell type (Fig. 6A,B). We next assessed cell-type composition in our tumor and benign mammary tissue samples by immunofluorescence (Fig. 6C). Although the immunopositive area was lower across all cell types in benign samples, mesenchymal and epithelial cells remained the dominant populations in both groups, indicating the absence of major shifts in the most abundant cell types (Fig. 6D). To integrate cell-type-specific OXPHOS mRNA expression (scRNA-seq) with cell-type abundance in our samples, we calculated a cell-type-weighted OXPHOS mRNA Index, reflecting each population's estimated contribution to Complex I-V transcript levels in our tissues. The OXPHOS mRNA Index (Fig. 6E) indicated that epithelial cells were likely the dominant contributors to OXPHOS transcripts in the present tumor samples. When these values were summed across complexes to generate a Total OXPHOS mRNA Score, epithelial cells remained the primary contributors (Fig. 6F). Altogether, these analyses suggest that our whole-tissue RNA-seq signatures are consistent with canonical HER2-driven signaling and indicate that the reduction in OXPHOS gene expression observed by bulk-tissue RNA-seq likely cannot be explained by a loss of high-OXPHOS epithelial cells or a gain in low-OXPHOS cell populations. Instead, these findings support a cell-intrinsic suppression of mitochondrial gene expression in HER2-driven tumors.

Mitochondrial respiratory capacity is increased in HER2-driven mammary tumors

Considering our transcriptomic and proteomic data strongly suggest HER2-driven mammary tumors display reductions in mitochondrial content, we next assessed whether this would manifest as a reduction in respiratory capacity by measuring mitochondrial respiratory flux rates. In permeabilized tumors and benign mammary tissue, we assessed basal/endogenous respiration and respiration with saturating carbohydrate (pyruvate/malate) and lipid (palmitoyl-carnitine) substrates. For each protocol, several biological replicates (individual tumors) were utilized, spanning a continuum of whole tumor wet weights/volumes (S-Fig. 5A,B). Since no strong correlations were found between tumor wet weight or volume and basal or maximal respiration supported by carbohydrate- or lipid-linked substrates (S-Fig. 5C-J), all data points were pooled. When normalizing respiration to tissue wet weight (Fig. 7A), tumors displayed significantly higher basal/endogenous respiration and higher respiration with subsequent additions of pyruvate, malate (CI-linked respiration), ADP (State III respiration), glutamate (CI-linked respiration also supported by alpha-ketoglutarate anaplerosis), and succinate (to maximize CII-

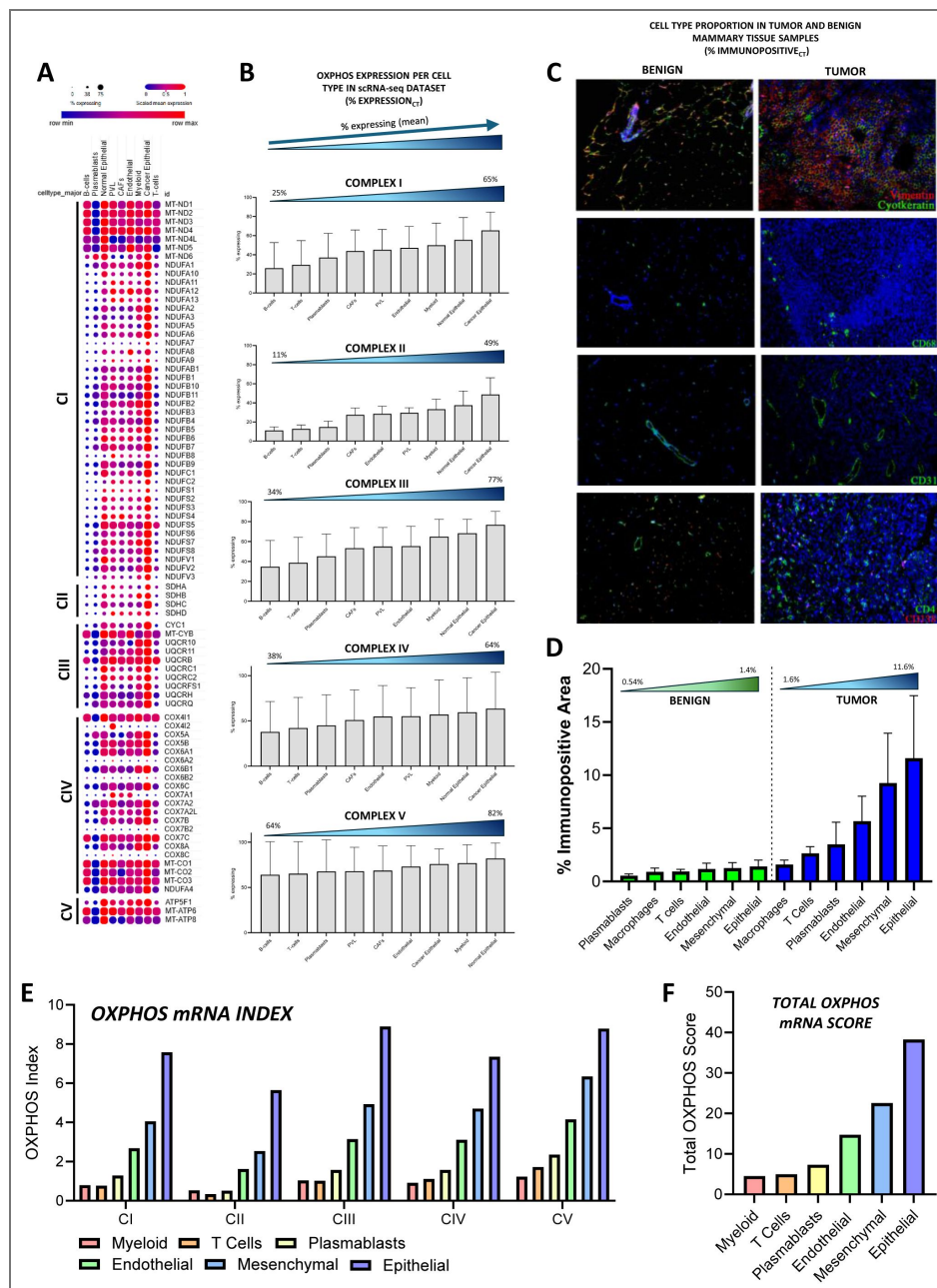


Figure 6. Cell type specific OXPHOS mRNA analysis in benign and tumor samples.

A) OXPHOS expression per cell type from a publicly available single-cell RNA sequencing dataset³¹ shown as a dot plot for each respiratory complex (Complexes I-V) subunit that was detected. Dot size indicates the percentage of cells of a given major type expressing each subunit gene (% cells expressing). Dot color represents gene expression scaled relative to each gene's expression distribution across all major cell types. **B)** Percentage of cells expressing OXPHOS subunits, arranged in ascending order per cell type for Complexes I-V. **C)** Representative immunofluorescence images of benign mammary tissue and tumor samples stained for vimentin, cytokeratin, CD68, CD31, CD4 and CD138. **D)** Quantification of cell-type proportion (% immunopositive area) in benign and tumor samples. **E)** OXPHOS mRNA Index calculated for Complexes I-V. $OXPHOS\ mRNA\ Index = (\%Expression_{CT}) * (\%Immunopositive_{CT})$ where CT represents cell type. **F)** Total OXPHOS mRNA Score for each cell type. $TOTAL\ OXPHOS\ mRNA\ Score = \sum_{i=I,II,III,IV,V} OXPHOS\ mRNA\ Index\ Ci$. Where i represents mitochondrial respiratory complexes.

linked respiration) (Fig.7B [↗](#)). Equally, tumors displayed significantly higher respiration rates supported by lipids (palmitoyl-carnitine, Fig.7C [↗](#)), suggesting that HER2-driven tumors are capable of relying on both substrate sources, perhaps to maximize aerobic metabolism. This initial analysis of respiratory rates was normalized to tissue wet weight, which can be influenced by differences in total and mitochondrial protein content in a given tissue volume. To address this, we first normalized respiratory flux rates to total cellular protein (Fig.7D,E [↗](#)), which minimizes differences in factors such as tissue density and hydration that are not directly contributing to mitochondrial oxygen consumption. The third normalization method employed was normalizing respiratory flux to a mitochondrial enrichment factor (MEF) (Fig.7F,G [↗](#)) which is the ratio of mitochondrial proteins to the underlying total proteome (S-Fig.3B [↗](#)). This approach controls for differences in mitochondrial content between tissues to determine whether increases in respiration are due to increases in mitochondrial content or intrinsic changes in mitochondrial function. Interestingly, the fundamental observation of increased respiratory flux rates in tumors compared to benign tissues was conserved independent of the normalization method, suggesting that HER2-driven mammary tumors are characterized by high respiratory flux rates independent of total cellular protein or reduced mitochondrial protein content. While greater intrinsic OXPHOS was observed with all substrates, lipid supported respiration was stimulated to a greater extent in the tumors, resulting in comparable carbohydrate (pyruvate, malate, ADP) and lipid (palmitoyl-carnitine, malate, ADP) supported respiration (Fig.8A [↗](#)). This effect occurred despite many of the genes and proteins involved in lipid metabolism at the plasma membrane and in the cytosol (e.g. *Acs1*, *Adipoq*, *Dgat1/2*, *Lipe*, *Fabp4*, *Cd36*, *Pparg*) and mitochondria (e.g. *Etf*, *Etfb*, *Etfdh*, *Hadh*, *Hadhb*, *Cpt1b*, *Cpt2*) being downregulated in tumors compared to benign mammary tissue (Fig.8B-E [↗](#)). The intrinsic activation of respiration was not a ubiquitous finding for all processes linked to OXPHOS, as reactive oxygen species (S-Fig.6A [↗](#)) emission rates were higher in tumors compared to benign tissue when normalized to wet weight (S-Fig.6B [↗](#)), lower when normalized to total protein (S-Fig.6C [↗](#)) and not different when normalized to mitochondrial content (S-Fig.6D [↗](#)).

High respiratory flux rates cannot be explained by OXPHOS uncoupling

Typically, respiratory capacity correlates with mitochondrial content in fully differentiated tissues³¹. However, given the apparent discrepancy between mitochondrial/OXPHOS protein content and respiratory flux rates in HER2-driven tumors, we investigated alternative explanations for this observation. One possible explanation for the observed enhanced respiratory rates could be due to the uncoupling of OXPHOS from ATP synthesis, especially when ROS emission is not proportional to respiratory flux. To address this possibility, we utilized oligomycin (OMY) to inhibit the F_0 subunit of ATP synthase (Complex V) and examine its inhibitory potency between groups (Fig.8F [↗](#)). While OMY inhibited respiration in benign mammary tissue by ~55%, respiration in tumors was inhibited by ~80% (Fig.8G [↗](#)). Subsequent additions of antimycin A (AMA) to inhibit Complex III and rotenone (ROT) to inhibit Complex I, fully inhibited respiration in both tissues (Fig.8G [↗](#)). Since respiration was inhibited to a greater extent in the tumors in response to OMY, this protocol strongly suggests that HER2-driven tumor mitochondria are highly coupled.

Tumor mitochondria display altered morphology

When assessing the subcellular ultrastructure of HER2-driven mammary tumors using transmission electron microscopy (TEM), we observed smaller, more punctate mitochondria in tumors compared to benign tissue (Fig.9A [↗](#)). To determine whether ultrastructural observations aligned with the expression of genes and proteins involved in mitochondrial turnover, we filtered our transcriptomic and proteomic dataset using MitoCarta, specifically investigating mitochondrial proteins involved in mitochondrial dynamics and mitophagy. This approach revealed the upregulation of several genes and proteins involved in mitochondrial fission and

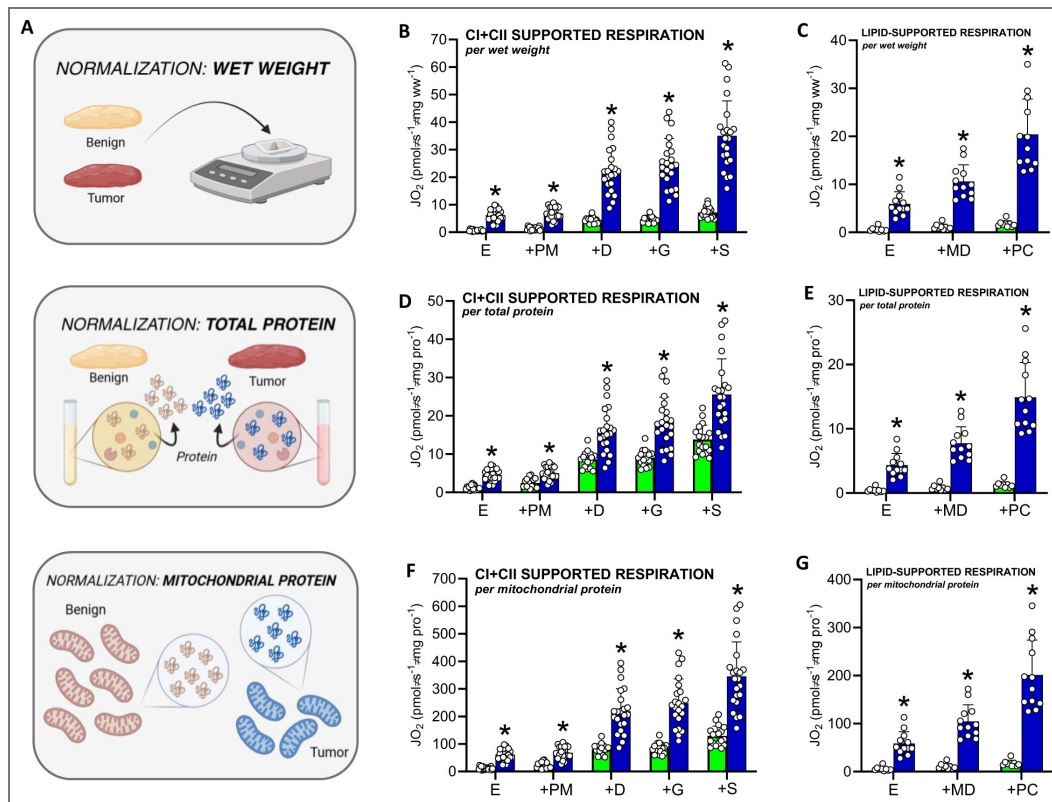


Figure 7. Assessment of lipid and carbohydrate linked mitochondrial respiration and OXPHOS coupling in benign mammary tissue and HER2-driven mammary tumors.

A) Schematic depicting methods of normalization for respiration rates (*per wet weight*, *total protein*, *mitochondrial protein/enrichment*). **B)** Complex I and complex II-supported respiration normalized to tissue wet weight, **C)** lipid-supported respiration normalized to tissue wet weight, **D)** Complex I and complex II-supported respiration normalized to total protein content, **E)** lipid-supported respiration normalized to total protein content, **F)** Complex I and complex II-supported respiration normalized to mitochondrial protein (mitochondrial enrichment factor, MEF). **G)** Lipid supported respiration normalized to mitochondrial protein. All individual values represent biological replicates (n=9-22/group depending on the protocol). Data are presented as means $\pm$ SD, statistical analyses were conducted using an unpaired, two-tailed Student's t-test (*p<0.05). Abbrev: E = Endogenous, PM = pyruvate (5 mM), malate (2 mM), D = ADP (5 mM), G = glutamate (10 mM), S = succinate (10 mM), PC = palmitoyl-carnitine (20 μ M), CI+CII = maximal CI+CII-supported respiration (PMDGS).

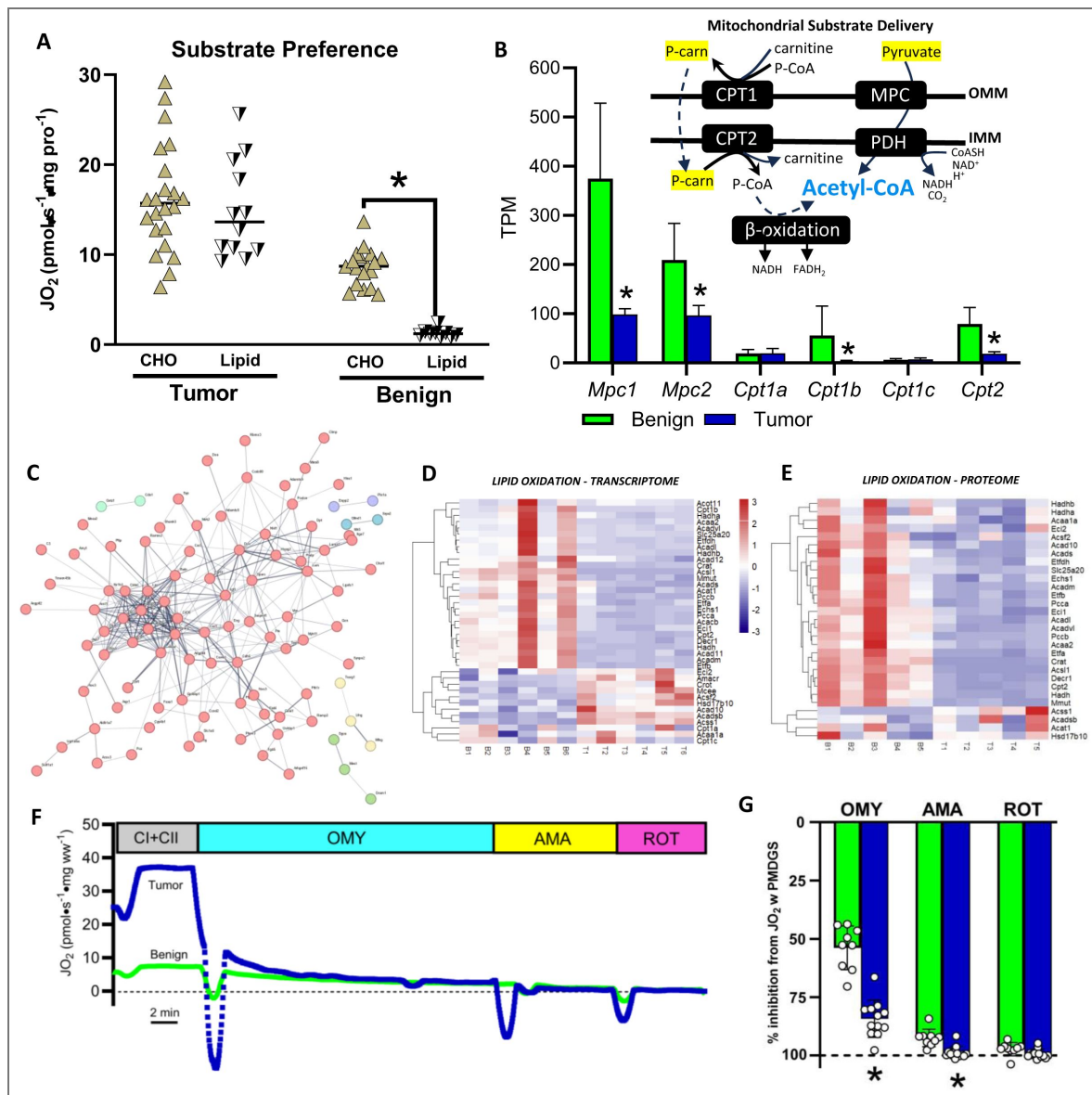


Figure 8. Lipid and carbohydrate linked respiration and lipid metabolism transcriptome and proteome.

A) Rates of carbohydrate (pyruvate, malate, ADP) and lipid (palmitoyl-carnitine, malate, ADP) supported respiration in tumors and benign mammary tissue. **B**) Gene expression (transcripts per million) of mitochondrial pyruvate carrier and carnitine palmitoyltransferase isoforms mediating mitochondrial pyruvate and long chain fatty acid transport respectively. **C**) Network interaction showing downregulated (unsupervised analysis) gene clustering in the HER2-driven tumor transcriptome, **D**) Heatmap of differentially expressed genes involved in lipid oxidation. **E**) Heatmap of differentially expressed proteins involved in lipid metabolism. **F**) Representative trace of protocol used to assess mitochondrial coupling. Following maximal rates of complex I and II- supported respiration, 1 μM oligomycin (OMY) was added to inhibit Complex V (ATP synthase), followed by 2.5 μM antimycin-A (AMA) and 1 μM rotenone (ROT) to inhibit complexes III and I respectively. **G**) Group data for experiment depicted in **F**) depicted as relative inhibition from maximal complex I and II supported respiration (PMDGS) in each group. In panels **A**, **B**, **F** and **G** data are presented as means $\pm$ SD and statistical analyses were conducted using an unpaired, two-tailed Student's t-test (* $p < 0.05$).

generally a downregulation of genes and proteins involved in mitochondrial fusion (Fig.9B-F [↗](#)). Together, these data show morphological differences in tumor versus benign tissue mitochondria, which could potentially influence OXPHOS efficiency.

HER2 inhibition supresses tumor respiration

Although the exact mechanism(s) driving OXPHOS activation despite reductions in mitochondrial content in our model remain unclear, we next asked whether HER2 activity directly regulates mitochondrial bioenergetics. To address this, we used NF639 cells—an epithelial line derived from mammary tumors from MMTV-c-neu transgenic mice, which, like our MMTV/neu^{nd1}-YD5 model, display upregulated murine HER2/neu signaling^{32–35} serving as a highly relevant cell-line analog to our intact tumor measurements. NF639 cells were treated with lapatinib, a dual HER2/EGFR tyrosine kinase inhibitor (Fig.10A [↗](#)) a well-described and validated inhibitor of HER2-mediated signaling^{29,36,37}. Cells were exposed to 250nM lapatinib for 24 hours, a dose and time regimen that was validated and did not notably compromise viability (Fig.10B [↗](#)). Using the same CI+CII linked respirometry protocol applied to the murine tumors (Fig.7B [↗](#)), lapatinib significantly reduced leak respiration, CI-linked respiration, and CI+CII-supported respiration relative to DMSO-treated cells (Fig.10C [↗](#)). Importantly, lapatinib did not induce a cytochrome c (CC) response (% change in respiration with CC, CON=−0.83%, LAP=−2.9%) and did not alter respiratory inhibition with OMY, AMA, or ROT (Fig.10C [↗](#)), indicating that mitochondrial integrity and coupling were maintained. Collectively, these findings demonstrate that upregulated HER2-signaling directly stimulates mitochondrial bioenergetics.

Discussion

The aim of the current study was to investigate the metabolic characteristics of a mouse model of HER2-driven mammary cancer compared to benign mammary tissue. Specifically, we focused on the bioenergetic adaptations that have occurred with HER2-driven tumorigenesis both with respect to mitochondrial function (respiration and ROS) and mitochondrial content (total mitochondrial proteome and OXPHOS complexes). This investigation revealed several findings: 1) HER2-driven mammary tumors exhibit high respiratory rates compared to benign mammary tissue, 2) tumors are capable of oxidizing lipids and pyruvate to a similar extent despite the downregulation of several genes involved in lipid catabolism, 3) high tumor respiration rates cannot be explained by higher mitochondrial content, and, 4) HER2 signaling regulates the upregulation of tumor respiration. These data highlight that HER2-driven mammary tumors exhibit high OXPHOS rates despite lower mitochondrial content, challenging the typical bioenergetic relationship in fully differentiated tissues.

Relationship between mitochondrial content and function in breast cancer

Studies simultaneously assessing mitochondrial content and mitochondrial respiration, and the contribution of these factors to breast cancer malignant transformation are scarce. While reductions in ETC subunit content/activity have been described in mitochondria isolated from human mammary carcinoma cultures compared to mammary epithelial cells³⁸, it has been assumed that a reduction in ETC subunit content would manifest as a reduction in respiratory capacity. However, this is at odds with several reports that classify breast cancer cells as OXPHOS reliant due to increases in mitochondrial respiration correlating with advanced disease^{19,39,40}. Indeed, across many cancers it is generally assumed that mitochondrial content and respiratory function increase or decrease concomitantly, however, recent work has challenged this dogma, demonstrating that high rates of respiration normalized to tissue weight can be negated when corrected for mitochondrial mass,^{41,42} revealing intrinsic alterations in cancer cell bioenergetics. In the current study, utilizing this same approach to control for differences in mitochondrial content, we found that respiratory capacity supported by pyruvate and lipids in HER2-driven mammary tumors was significantly higher (~2–4 fold) compared to benign mammary tissue despite lower mitochondrial enrichment (total mitochondrial proteome) and lower abundance of

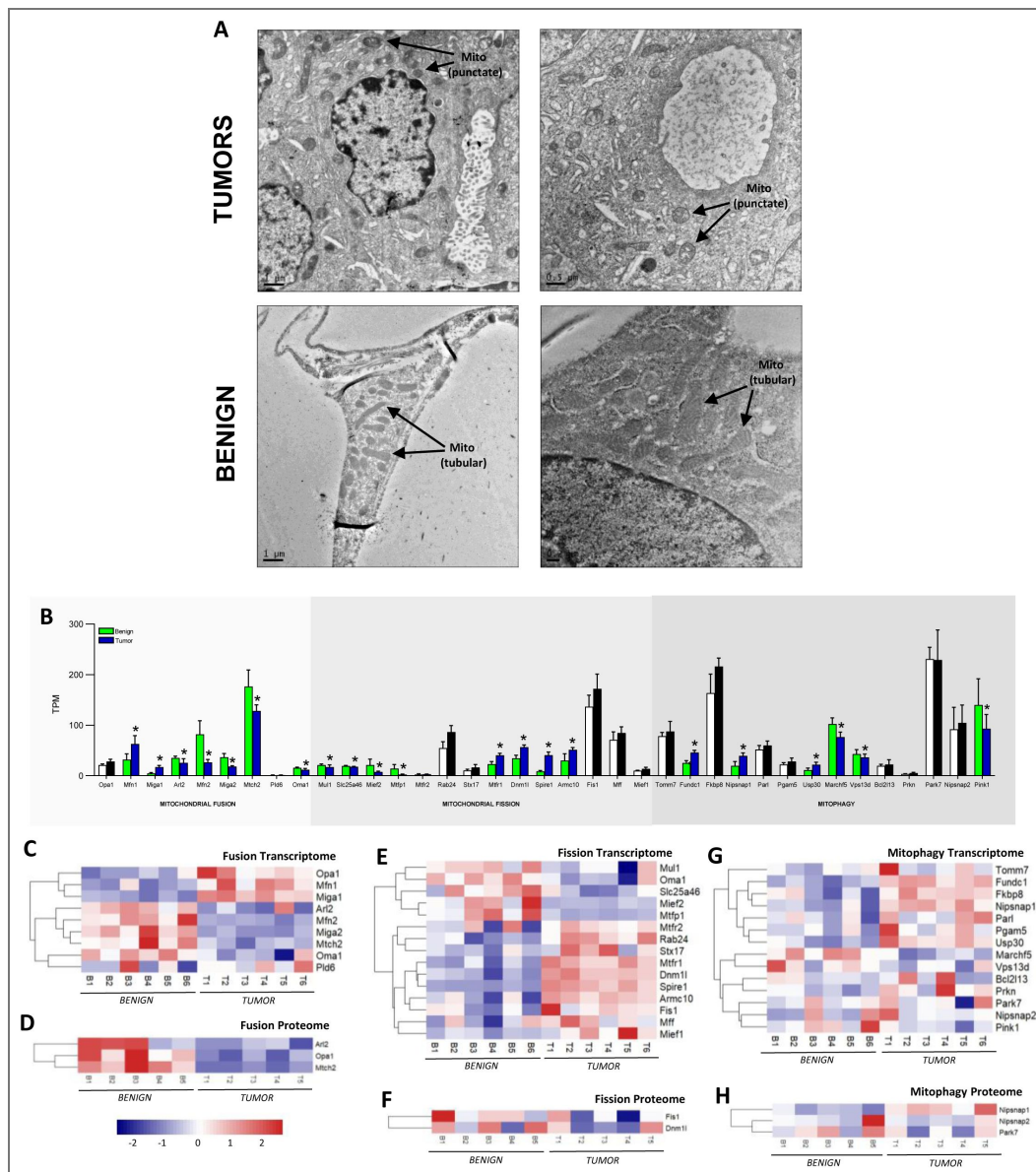


Figure 9. Mitochondrial morphology and assessment of mitochondrial fusion, mitochondrial fission and mitophagy transcriptomes and proteomes in benign mammary tissue and HER2-driven tumors.

A) Transmission electron microscopy (TEM) images of tumor and benign samples showing punctate (tumor) versus elongated (benign) mitochondria. **B)** Gene expression (detected by transcriptomics; transcripts per million) of fusion, fission and mitophagy related genes separated per process filtered by MitoCarta, coloured bars and * represent genes with an adjusted p value of <0.1. Black and white bars represent genes with no significant differences between groups. **C)** Heatmap of fusion transcriptome, **D)** Heatmap of fusion proteome, **E)** Heatmap of fission transcriptome, **F)** Heatmap of fission proteome, **G)** Heatmap of mitophagy transcriptome, **H)** Heatmap of mitophagy proteome. TEM images are from independent tumor and benign tissue samples. Heatmaps depict all detected proteins (n=5/group) and transcripts (n=6/group).

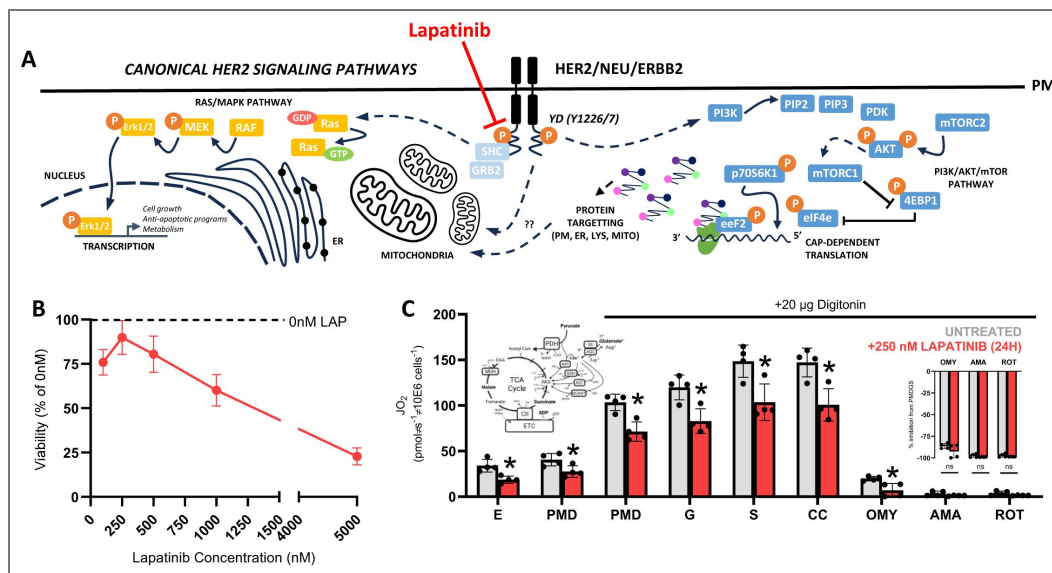


Figure 10. Mitochondrial respiration is suppressed by tyrosine kinase inhibitor lapatinib.

A) Schematic showing mechanism of lapatinib mediated inhibition of HER2 tyrosine kinase activity. **B)** NF639 cell viability assessed using a crystal violet assay expressed as a percentage of untreated viability (0nM lapatinib). **C)** Complex I and II supported respiration in the presence of 20μg digitonin in NF639 cells untreated (grey) or treated with 250 nM lapatinib for 24h. Left inset: Schematic of substrates supporting Complex I-II respiration, schematic of lapatinib or DMSO treatment. Right inset: Percent inhibition of maximal CI+II supported respiration (PMDGSC) with OMY, AMA and ROT. Following maximal rates of complex I and II- supported respiration, 1 μM oligomycin (OMY) was added to inhibit Complex V (ATP synthase), followed by 2.5 μM antimycin-A (AMA) and 1 μM rotenone (ROT) to inhibit complexes III and I respectively. Each datapoint represents a technical replicate (duplicate) and data are presented as means ± SD. Statistical analyses were conducted using an unpaired, two-tailed Student's t-test (*p<0.05). **Abbrev:** E = Endogenous, PMD = pyruvate (5 mM), malate (2 mM), D= ADP (5 mM), G = glutamate (10 mM), S = succinate (10 mM), CC = cytochrome C (10 μM).

several respiratory complex subunits, suggesting mitochondrial content independent increases in mitochondrial respiratory capacity are acquired during HER2-driven tumorigenesis. This is a key finding independent of the specific mechanisms underlying the enhanced bioenergetics.

While the exact mechanisms underlying the activation of OXPHOS in the current work remain unclear, our findings allow us to speculate on several possibilities that could enhance respiration in a mitochondrial-content-independent manner. It is unlikely these factors are mutually exclusive and include increased coupling of electron transfer to ATP synthesis (coupling efficiency), changes in the delivery of substrates across the mitochondrial membranes, intrinsic changes in the activity of ETC complexes and/or ATP synthase, alterations in mitochondrial morphology/dynamics and signaling events linked to HER2 activation.

Possible mitochondrial content-independent mechanisms supporting enhanced respiration rates in HER2-driven mammary cancer

Coupling Efficiency

Several studies have suggested that a key function of mitochondrial respiration in cancer cells is the regeneration of NAD^+ , FAD and ubiquinone to provide the necessary metabolites for biomass production, rather than ATP synthesis^{5,6,43}. From a bioenergetic perspective, this could be achieved by inducing uncoupled respiration, which could manifest in an apparent increase in intrinsic mitochondrial respiration, given that measures of respiratory flux quantify oxygen consumption rather than ATP synthesis. However, in the present study the Complex V inhibitor oligomycin inhibited respiration to a greater extent in HER2-driven tumors, suggesting that uncoupling cannot explain the enhanced respiration rates observed, and if anything, tumors are more coupled than benign mammary tissue. This interpretation aligns with work in MCF7 cells where mitochondrial ATP represents the majority of cellular ATP production⁴⁴ suggesting the ATP synthetic function of OXPHOS in breast cancer is likely intact. Indeed, enhanced mitochondrial ATP provision and OXPHOS-dependent biomass expansion are not mutually exclusive processes and increased coupled respiration would satisfy both the bioenergetic and biosynthetic requirements of rapid cell proliferation during tumorigenesis.

Substrate Delivery and Preference

Recent work has highlighted the dependency of HER2+ tumor cells on both lipid (long chain fatty acids) and carbohydrate-derived (pyruvate) substrate for optimal proliferative capacity⁴⁵. In addition to pyruvate, HER2-driven tumors may be particularly poised to utilize lipids, given the proximity of cancer cells to adipocytes in the mammary tumor microenvironment⁴⁶, which has spurred interest in inhibiting lipid use/fat oxidation as an adjunct therapy in HER2-driven cancer⁴⁵. Indeed, the use of endogenous substrate may be preferred as solid tumors expand and factors such as increased interstitial fluid pressure and extracellular matrix remodelling impact tumor vessel quality⁴⁷, possibly limiting exogenous substrate delivery. However, at the transcript and protein level, enzymes involved in mitochondrial lipid uptake (CPT1, CPT2) and pyruvate uptake (MPC) were largely downregulated compared to benign tissue which would limit rather than enhance mitochondrial substrate delivery. Despite this lower abundance of enzymes supporting mitochondrial substrate delivery, mitochondrial respiration supported by pyruvate and lipids were significantly higher in tumors. Considering lipid and carbohydrate oxidation converge within the mitochondrial matrix, both providing acetyl-CoA to the TCA cycle, it is possible that enhanced OXPHOS efficiency lies at a distal site, implicating intrinsic OXPHOS adaptations as a possible explanation for enhanced respiration in HER2-driven tumors.

Intrinsic OXPHOS Adaptations

Several cancers are characterized by the altered expression of individual ETC subunits and/or ATP synthase which can have functional repercussions on respiratory function^{10,41}. In the current study, despite a consistent downregulation of subunits comprising Complexes I-IV in HER2-driven tumors, a subset of Complex V genes were not downregulated (specifically two subunits of the c-

ring of F_0 domain (*Atp5g1*, *Atp5g2*), two subunits of the F_6 domain stalk connecting the F_0 and F_1 domains (*Atp5j1*, *Atp5j2*), and regulatory proteins ATPase inhibitory factor 1 (*Atpif1*) and distal membrane assembly component 2 (*Dmac2l*). This shift essentially increases the ratio of ATP synthase/ETC and raises the possibility that a maintenance of these subunits may enhance OXPHOS rates in the face of downregulated CI-IV subunit content since maximal coupled respiration is usually limited by proton flux through ATP synthase. Interestingly this observation was further supported by our assessment of scRNA-seq data in human breast tumors, where CV expression was generally uniformly and highly expressed across major cell types, further supporting the notion that the maintenance of CV may be important for tumorigenesis. Indeed, despite mtDNA lesions in proton pumping complexes (CI, CIII and CIV) being common in many cancers, presumably reducing the content/function of these complexes, ATP synthase mutations are less common, suggesting positive selection may exist to maintain coupled respiration and ATP production^{10,48,49}. Thus, while speculative, it is possible that higher respiration rates in HER2-driven tumors may partially reflect maintained ATP synthase content/function.

While bioenergetic efficiency is typically related to intrinsic changes in OXPHOS stoichiometry, mitochondrial morphology can indirectly affect efficiency. We observed smaller, punctate mitochondria in HER2-driven mammary tumors when visualized by TEM, generally aligning with reductions in mitochondrial fusion and increases in mitochondrial fission-related targets. While the direct relationship between mitochondrial morphology and bioenergetics remains contentious, evidence suggests smaller mitochondria (due to enhanced fission or reduced fusion) may be associated with enhanced bioenergetics. Smaller mitochondria tend to have a higher surface-area-to-volume ratio which has been correlated with upregulated OXPHOS in the liver⁵⁰, and low rates of respiration is associated with the inhibition of mitochondrial fission⁵¹. Furthermore, rare genetic mitochondrial diseases are often characterized by hyper-fused mitochondria which is associated with OXPHOS limitations⁵². Indeed, morphological differences in breast cancer mitochondria are well described^{16,53–55} and targeting mitochondrial dynamics for therapy is an active area of study^{56–58}, however, there is a clear need to ascertain the relationship between mitochondrial dynamics and bioenergetics in tumorigenesis.

Intrinsic activation of OXPHOS mediated by HER2

No matter the mechanisms underlying our observed bioenergetic activation in HER2-driven tumors, our experiments in NF639 cells demonstrate that HER2 activation appears to regulate respiratory function. Although findings in the literature regarding how HER2 activation or suppression influences mitochondrial content and bioenergetics remain mixed, several lines of evidence support the idea that HER2 is important for tumorigenesis and respiratory competence^{19,59–61}. Importantly, these studies collectively reinforce that changes in mitochondrial content alone cannot fully account for alterations in respiratory function, suggesting that HER2 may regulate intrinsic mitochondrial activity. Independent groups have shown that HER2 may localize to the inner mitochondrial membrane and directly interact with respiratory complexes (CI,CIV,CV) influencing their activity^{19,59,60}. Alternatively, indirect activation of bioenergetics via HER2 signaling is another possibility – which is supported by work showing mitochondrial ADP availability may be enhanced via mitochondrial creatine kinase (MtCK) phosphorylation by HER2 downstream effectors⁶¹. Both mechanisms would alter bioenergetics without necessarily altering mitochondrial content, which aligns with our observation that content markers do not scale with respiratory capacity. Nonetheless, further research is required to more clearly delineate the relationship between HER2 signaling, OXPHOS activation and possible mitochondrial content independent mechanisms conferring respiratory competence.

Conclusion

We provide evidence that HER2-driven mammary tumors display enhanced respiratory capacity despite lower mitochondrial content. This increased respiratory capacity could not be explained by changes in OXPHOS coupling, however, differences in mitochondrial morphology and the maintained expression of several subunits of ATP synthase suggest HER2-driven mammary tumors

may undergo intrinsic adaptations in mitochondrial function during tumorigenesis. No matter the mechanism driving OXPHOS activation in our model, upregulated HER2 signaling is essential for the observed increases in mitochondrial respiratory capacity.

Materials and methods

Animals and Breeding

All rodent experiments were conducted in accordance with institutional guidelines approved by the Animal Care Committee at the University of Guelph (AUP#5071). MMTV/*neu*^{ndl}-YD5 mice on an FVB background were obtained from Dr. William Muller (McGill University) and Dr. David Ma (University of Guelph). Female MMTV/*neu*^{ndl}-YD5 mice were obtained by breeding male heterozygous MMTV/*neu*^{ndl}-YD5 mice with female FVB mice to yield WT or heterozygous transgenics. Mice were group housed in ventilated cages at room temperature (22°C) and kept on a 12:12 light-dark cycle with *ad libitum* access to a chow diet and water. Starting at 10 weeks, mice were monitored for tumor development on a biweekly basis for ~10 additional weeks. Mice with tumors exceeding 17 mm in length/width or a volume over 5000 mm³ were euthanized before the 20-week time point.

Tumor and Benign Mammary Tissue Collection

Mice were anesthetized using sodium pentobarbital (60 mg/kg body weight) and checked for depth of anaesthesia by leg retraction after tail pinch. Once under the surgical plane, a ventral incision was made up the midline and the skin separated bilaterally from the underlying fascia to allow for the excision of the intact mammary tumors. Depending on the number of tumors present (typically 1-5), tumors were allocated for mitochondrial function (placed fresh in BIOS preservation buffer), histology (TEM: fixed in 2.5% glutaraldehyde/1% paraformaldehyde, IF: fixed in 10% formalin) or snap frozen in liquid nitrogen for further analysis (western blotting, transcriptomics, quantitative label-free proteomics) immediately after whole tumor wet weight was recorded. Paired mammary tissue with no visible tumors, furthest from the site of tumor excision (i.e. if tumors localized in cervical region benign tissue was harvested from inguinal region) were used as an internal comparison, denoted as benign mammary tissue. While mitochondrial respiration experiments were performed in paired tissue (benign and tumor from same mouse), transcriptomic, proteomic and histological analyses were performed on tumors and benign samples from different mice due to tissue limitations.

RNA Sequencing and Transcriptomic Analyses

Frozen tissues were homogenized and lysed in TRIzol reagent. After centrifuging, the supernatant (aqueous phase) was applied to the RNeasy kit (Qiagen, 74106) for subsequent total RNA extraction and purification in accordance with the manufacturer's protocol. RNA samples were sent to the McMaster Genomics Facility where an RNA quality check, poly A enrichment, library prep and quality check were performed as previously described⁶². These samples were then run in an Illumina Nextseq P2, 2×50bp sequencing run. FastQC and MultiQC were used for quality control of raw data. Salmon's transcript-level quantification⁶³ and DESeq2⁶⁴ was used to detect DEGs with the threshold of adjusted p-value<0.1. PCA analysis was performed by using variance stabilizing transformation (VST) data through DESeq2 using the pcaExplorer package⁶⁵ in R (version 4.2.1). Functional enrichment analysis was performed by GO enrichment analysis⁶⁶ and Kyoto Encyclopedia of Genes and Genomes (KEGG) mapping using the GOSTats, KEGG.db, and Bioconductor packages. Heatmaps and volcano plots were generated with ggplot2 package using all detected genes or genes sorted by an adjusted p-value<0.1 by false discovery rate method. Detailed descriptions for each analysis are included in the figure descriptions.

Western Blotting

Tumor and benign mammary tissues were homogenized in lysis buffer, diluted to equal protein concentration (1 µg/µL), separated by SDS-PAGE, transferred onto PVDF membranes and detected using enhanced chemiluminescence (ChemiGenius2 Bioimaging System, SynGene, Cambridge, UK)

as previously described^{67,68}. Commercially available primary antibodies were used to detect total and phosphorylated targets (p38 MAPK (CS9212S, 1:1000), ERK1/2 (CS9102S, 1:1000), Akt (CS4691S, 1:1000), mTOR (CS2972S, 1:1000), p70s6k1 (CS9202S, 1:1000), eEf2 (CS2332S, 1:1000), p-p38 MAPK (CS9211S, 1:1000), p-ERK1/2 (CS9101S, 1:1000), p-AKT (CS9271S, 1:1000), p-mTOR (CS2971S, 1:1000), p-p70s6k1 (CS9234S, 1:1000), p-eEF2 (CS2331S, 1:1000), OXPHOS complexes CII and CIV (Ab110413, 1:1000)). Due to rapid tissue/protein turnover in tumor tissues, Ponceau staining was an inappropriate loading control, however, trends in total protein targets between groups aligned with quantitative label-free proteomic analyses as a secondary method measuring protein content, validating targets detected by western blotting.

Protein Extraction and Purification for LC-MS

A Bradford assay was used to determine the protein concentration in the samples. Twenty-five µg of protein per lysate was resolubilized with denaturation buffer (6M urea/2M thiourea) and concurrently reduced with 10 mM DTT and alkylated with 20 mM iodoacetamide at RT for 60 min. Samples were precipitated by adding 6:1 v/v cold acetone and kept at -80°C for 60 minutes and the pellet was collected after centrifugation at 10,000g for 10 min at 4°C. Samples were resuspended with 50 mM ammonium bicarbonate and MD-grade trypsin (Thermo Fisher Scientific, Cat#90057) was added at a ratio of 1:50 protease to protein. Digestion occurred overnight at 37°C. After digestion, samples were dried by vacuum centrifugation and purified using Pierce C18 Spin columns (Thermo Fisher Scientific, cat#89873).

LC-MS

The Vanquish Neo UHPLC system was coupled with Orbitrap Exploris 240 mass-spectrometer using the Easy-Spray source for nanoLC-MS protein identification. The Vanquish Neo UHPLC system was configured for trap and elute analysis. Peptides were first trapped and washed on a Pepmap Neo C18 trap column (5 micron, 300 micron x5 mm) then separated on EASY-Spray columns 75 µm I.D. × 50 cm with the maximum pressure of 1200 bar. The nanoLC-MS system was controlled with Standard Instrument Integration (SII) for Xcalibur software. All hardware and data acquisition software were from Thermo Fisher Scientific. The mobile phase A and weak wash liquid was water with 0.1% FA, and the mobile phase B and strong wash liquid was 80% acetonitrile with 0.1% FA. The gradient was as follows 4-19% B over 72 min, 19-29% B over 28 min, 29-45% B over 20 min and a 14.5 min wash at 100% B with a flow rate of 300 nL/min. The autosampler temperature was 7 °C, and the column temperature was 45 °C. The sample was injected with Fast Loading set to 'Enabled' with Pressure Control at 500 bar. The column Fast Equilibration function was set to 'Enabled' with Pressure control at 800 bar, and the equilibration factor was set to 3. Vial bottom detection was set to 'Enabled'.

DDA MS Method

The Orbitrap Exploris 240 MS was operated in DDA mode using a full scan with m/z range 375–1500, Orbitrap resolution of 60 000, normalized AGC target value 300%, and maximum injection time set to Auto. The intensity threshold for precursor was set to 1×10^4 . MS/MS spectra starting from 120 m/z were acquired in data-dependent acquisition (DDA) mode with a cycle time of 2 s, where the precursors were isolated in a window of 1.6 Da and subsequently fragmented with HCD using a normalized collision energy of 30%. Orbitrap resolution was set to 15 000 for MS2. The normalized AGC target was standard, and the maximum injection time was set to Auto.

Proteomics data analysis

Analysis of the tissue proteome was performed in R (version 4.2.1) and R Studio (version 2024.04.2+764). Principal component analysis was performed using the Vegan package. Volcano plots were created using log2 transformed fold change comparing cancer/benign tissue and -log10 adjusted p-value. P-values of differentially regulated proteins between each group were corrected with p.adjust (method = "fdr") within each comparison. Depending on the analysis, heatmaps and volcano plots were generated using all detected proteins or proteins sorted by p.adjust <0.1,

detailed descriptions for each analysis are included in the figure descriptions. Gene set enrichment was performed with the Kyoto Encyclopedia of genes and genomes (KEGG)⁶⁹ and EnrichR⁷⁰.

Cell Culture

NF639 murine epithelial mammary tumor cells (ATCC: CRL-3090) were cultured in DMEM supplemented with 10% fetal bovine serum (FBS; Gibco), 4 mM L-glutamine, 1 mM sodium pyruvate, 50 U/mL penicillin and 50 ug/mL streptomycin (Wisent), maintained by passaging in 100 mm plates, and incubated at 37°C and 5% CO₂. Cells were seeded in 96-well plates (10⁴ cells/well; n=3 technical replicates) for microplate viability assays or in 100 mm plates for mitochondrial respiration experiments. At 60% confluency, cells were serum-starved for 6 hours, followed by lapatinib treatment (Cayman, 11493) in complete culture medium containing 2% FBS for 24 hours. Cell viability in the presence of lapatinib was tested using a crystal violet assay. Cells were treated with lapatinib at 0 nM, 100 nM, 250 nM, 500 nM, 1000 nM, or 5000 nM for 24 hours. Following treatment incubation, cells were washed and stained with crystal violet dye (Abcam) and absorbance was measured at 570 nm as per the manufacturer's instructions. Assays were repeated in triplicate. For mitochondrial respiration experiments, cells were treated with 250 nM lapatinib or DMSO (control) in complete culture medium containing 2% FBS for 24 hours. Cells were washed, trypsinized (1x trypsin-EDTA, Wisent), and centrifuged at 300g for 5 min before counting. Cells were washed and resuspended in 1x PBS and centrifuged at 300g for 5 minutes. Cells were resuspended in MirO5 respiration buffer (0.5 mM EGTA, 3 mM MgCl₂·H₂O, 60 mM potassium lactobionate, 10 mM KH₂PO₄, 20 mM HEPES, 110 mM sucrose, 20 mM taurine, and 1 g/L fatty acid free BSA; pH 7.1) for mitochondrial respiration.

Mitochondrial Respiration

Mitochondrial respiration experiments were performed using the Oroboros Oxygraph-2k systems at 37°C in MirO5 respiration with constant stirring at room air saturation (starting at 180 μM O₂). For respiration experiments in intact tumors and benign tissue, 2-5 mg of permeabilized tumor tissue or 15-20 mg of permeabilized (both permeabilized with 20 μg/μL saponin) mammary tissue were used per 2 mL chamber and each protocol was conducted as at least two technical replicates (duplicate) per tumor or benign sample. Prior to the addition of substrates for any protocol, endogenous (basal) respiration was assessed with no substrates added. Complex I and Complex II supported respiration was assessed with the addition of 5 mM pyruvate and 2 mM malate, 5 mM ADP, 10 mM glutamate and 10 mM succinate. This respiratory capacity measurement represents supraphysiological substrate concentrations delivered in the absence of diffusion limitations. In a subset of these experiments, subsequent additions of oligomycin (OMY, 1 μM), antimycin A (AMA, 2.5 μM) and rotenone (ROT, 1 μM) were added to assess OXPHOS coupling. In a separate protocol, lipid-supported respiration was assessed with 20 μM palmitoyl-carnitine in the presence of 2 mM malate and 5 mM ADP. For respiration experiments in NF639 cells, the same respiration buffers and chamber conditions were utilized. One million cells were added per technical replicate and Complex I and II supported respiration was assessed using the same protocol described in intact tumors and benign tissue in the presence of 20 μg digitonin to ensure permeabilization. Respiration was expressed per viable cell.

Reactive Oxygen Species Emission

Mitochondrial hydrogen peroxide (H₂O₂) emission was determined fluorometrically (Lumina, Thermo Scientific, Waltham, MA) in permeabilized mammary tissue and mammary tumors as previously described⁶⁷ with minor modifications. In brief, 2-5 mg of minced tissue was loaded into a cuvette containing Amplex Red (Invitrogen, Waltham, MA, USA), horseradish peroxidase (5U/mL), saponin (20 μg/μL), superoxide dismutase (40 U/mL) in Buffer Z (105 mM K-MES, 30 mM KCl, 1 mM EGTA, 10 mM K₂HPO₄, 5 mM MgCl₂, 5 μM glutamate, 5 μM malate, 0.5% BSA, pH 7.4). Emission rates were assessed at 37°C in the presence of 10 mM succinate and calculated using a standard curve generated with known concentrations of H₂O₂.

Data Normalization

Mitochondrial respiration rates and H_2O_2 emission were normalized to individual sample wet weight, group total protein content and group mitochondrial content (mitochondrial enrichment factor, MEF). Total protein content was calculated as a group average ($n=6/\text{group}$) based on a Bradford assay and back-calculated total protein in a given wet weight of starting tissue. Data was normalized to mitochondrial protein by calculating the MEF for each group ($n=g/\text{group}$) based on all detected mitochondrial proteins by quantitative label-free proteomics (Mouse MitoCarta 3.0 database) normalized to total protein abundance per sample as previously described by others^{31,42}.

Histology

Transmission Electron Microscopy

Immediately following tissue dissection, tumor and benign mammary tissues were fixed in 2.5% glutaraldehyde/1% paraformaldehyde overnight at 4°C. The following day, tissues were washed 3x in 100 mM HEPES buffer in preparation for embedding. Tissue was stained with 1% osmium tetroxide for 3h, washed 3x in HEPES buffer and further stained with 1% uranyl acetate for 3h before washing 2x in HEPES buffer and once in ddH₂O. Stained tissue was dehydrated with a graded ethanol series followed by infiltration with LR White resin. Blocks were polymerized at 60°C for 18 hours before ultramicrotomy. Sections were post-stained with uranyl acetate and lead citrate. Sections were allowed to fully dry before imaging on the FEI Tecnai G2 F20 transmission electron microscope operated under normal conditions at 120 kVe.

Immunofluorescence

Tumor and benign samples ($n=4/\text{group}$) were fixed in 10% formalin for 24 hours and stored in 70% ethanol at 4°C. Samples were blocked in paraffin and sectioned, deparaffinized in xylene, and rehydrated through a graded ethanol series. Samples were washed in PBS and incubated for 20 min in 0.02% sodium borohydride in PBS to quench endogenous phosphatase activity. Antigen retrieval was performed in citrate buffer (90°C) for 12 min, followed by a 20-min cooling period. Sections were then blocked in 5% BSA in PBS for 1h to prevent nonspecific antibody binding. Primary antibodies (anti-vimentin (1:100, Novus Bio NBP1-31327), anti-cytokeratin (1:100, Novus Bio NBP2-29429), anti-CD4 (1:200, Abcam AB183685), anti-CD138 (1:100, StemCell Technologies 60035), anti-CD31 (1:150, Abcam AB28364), or anti-CD68 (1:100, Novus Bio NB100-683)) were diluted in 1% BSA and applied overnight at 4 °C in a humidified chamber. The following day, sections were incubated for 2 hours at room temperature with the appropriate fluorophore-conjugated secondary antibodies (Invitrogen): Alexa Fluor 594 anti-rabbit (1:100 or 1:200), Alexa Fluor 488 anti-mouse (1:100), Alexa Fluor 488 anti-rat (1:200), or Alexa Fluor 488 anti-rabbit (1:100). Slides were washed and mounted with ProLong™ Gold Antifade Mountant containing DAPI (Thermo Fisher) and protected from light for 24 hours before being transferred to -20°C for storage. Images were acquired at 40× magnification using an Olympus BX-61 epifluorescence microscope equipped with Metamorph imaging software. Monochromatic images captured at 594 nm excitation were pseudocolored red, while those captured at 488 nm were pseudocolored green. Staining intensity was quantified in ImageJ using manual region-of-interest selection, and immunopositive area was expressed as a percentage of total tissue area (expressed as % immunopositive area).

OXPHOS Index and OXPHOS Score Calculations

A publicly available single-cell RNA-seq dataset from 26 human breast tumors³¹ was combined with cell-type proportion analyses from our tumor and benign tissues (see Immunofluorescence methods) to estimate cell-type-specific contribution to OXPHOS.

Using human MitoCarta 3.0, we filtered for all detected OXPHOS subunit genes in the scRNA-seq dataset. For each major cell type described in the dataset (B-, T-, plasmablasts, CAFs, PVL, endothelial, myeloid, normal epithelial, and cancer epithelial cells), we extracted the percent of cells expressing each subunit gene. These percent expression values were averaged per complex to generate a mean CI-V expression value for each cell type. To account for differences in cell-type

abundance within the scRNA-seq dataset, this percent expression was multiplied by the cell-type proportion reported in the same study. In subsequent calculations, this is reported as $\%Expression_{CT}$.

To estimate cell-type-specific OXPHOS expression in our benign and tumor samples, we calculated an *OXPHOS mRNA Index* for all cell types present in both datasets (myeloid, T-, plasmablasts, endothelial, mesenchymal, and epithelial cells):

$$OXPHOS\ mRNA\ Index = (\%Expression_{CT}) * (\%Immunopositive_{CT})$$

Where, $\%Expression_{CT}$ is the average gene expression data per respiratory complex in each complementary major cell type (CT), and $\%Immunopositive_{CT}$ is cell type proportion measured by IF in our samples. This index represents how much each cell type is expected to contribute to CI-V expression in our tissues, integrating both expression and abundance.

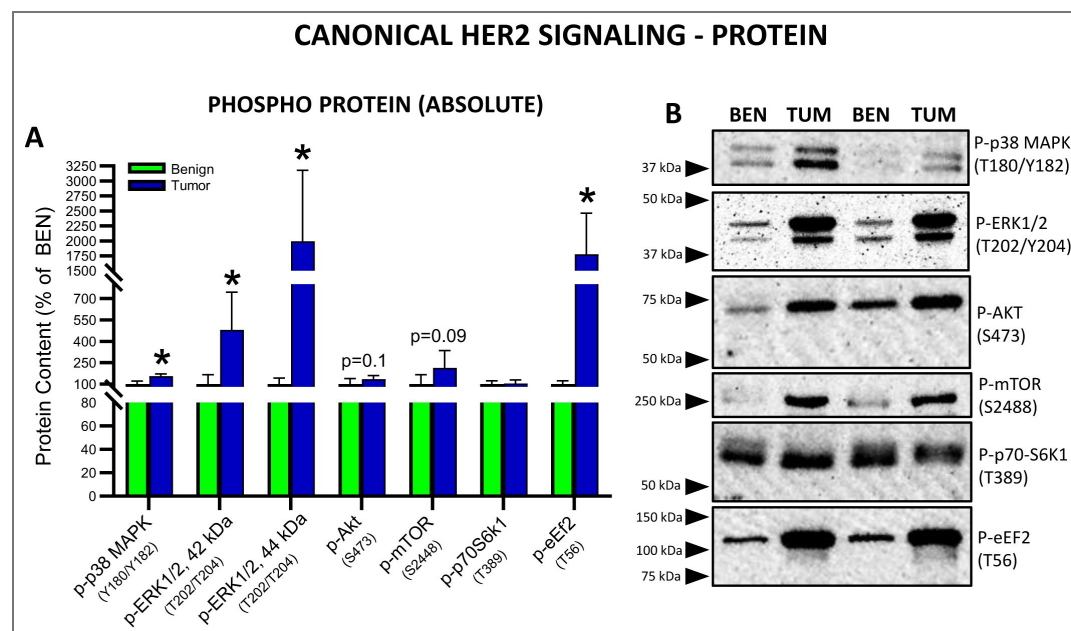
Next, to estimate the contribution of each cell type to the OXPHOS transcriptome in the current study, the OXPHOS mRNA index in each cell type were summed to calculate a Total OXPHOS mRNA Score.

$$Total\ OXPHOS\ mRNA\ Score = \sum_{i=I,II,III,IV,V} OXPHOS\ RNA\ Index\ C_i$$

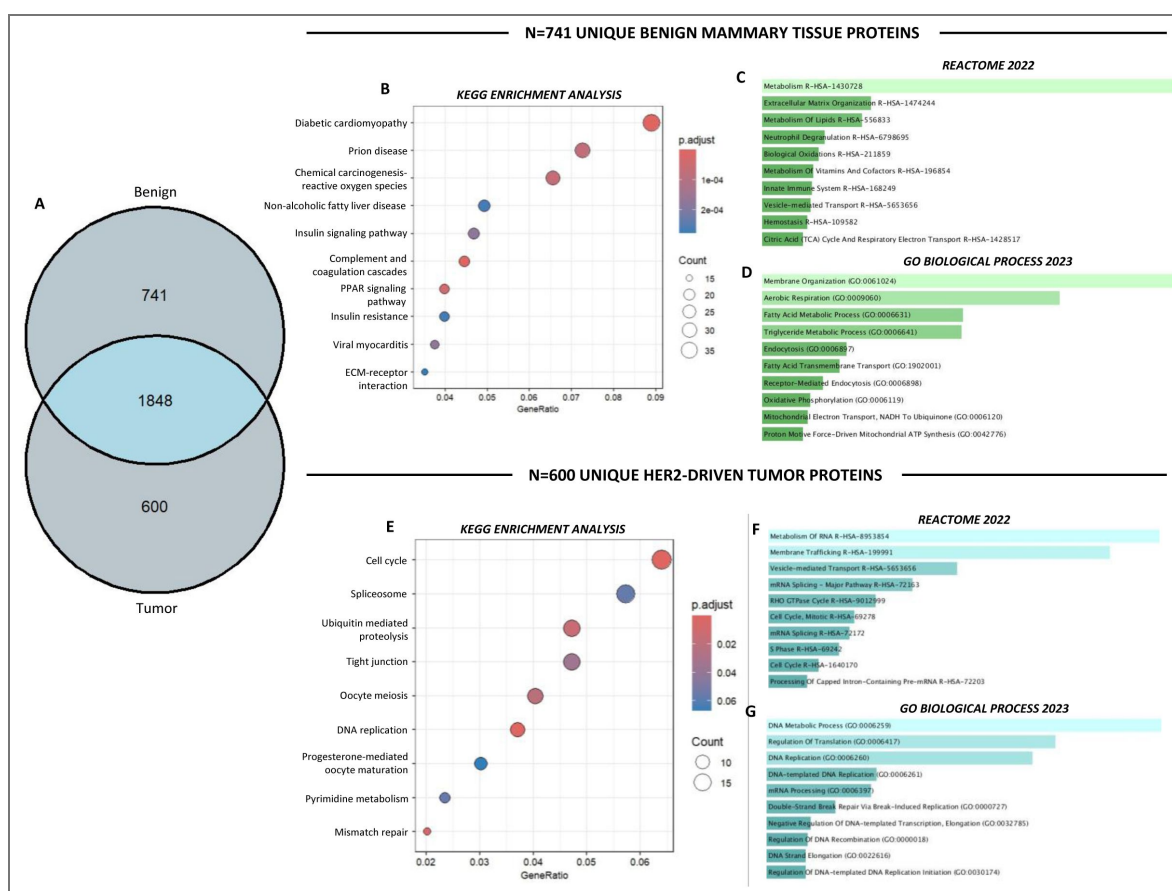
Statistical Analysis

Statistical analysis and data visualization was performed using Microsoft Excel, GraphPad Prism (version 10.2.3), R (version 4.2.1) and R Studio (version 2024.04.2+764). All data are presented as mean $\pm$ SD and data was considered statistically significant if $p < 0.05$. For transcriptomic and proteomic analyses, an adjusted p-value of < 0.1 was utilized. Details regarding specific statistical analyses are included in the Results section and within each figure legend.

Supplementary material

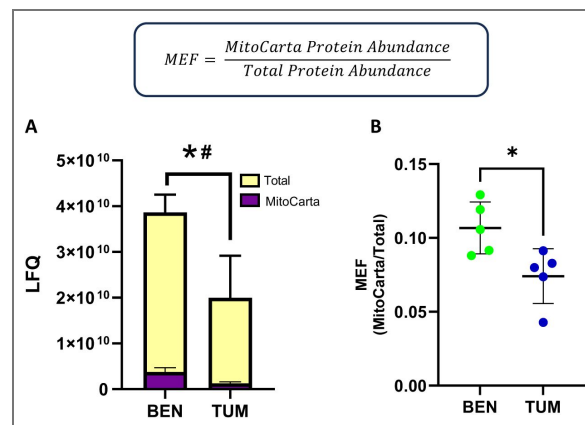


Supplementary Figure 1. Absolute phosphorylated protein targets downstream of HER2 activation. A) Quantified western blots for phosphorylated canonical HER2 signaling proteins (n=5-6/group). Phosphorylation sites are indicated for each target. **B)** Representative blot for each phosphorylated target. Data are presented as mean $\pm$ SD and analyzed with an unpaired, two tailed Student's t-test (* $p < 0.05$).



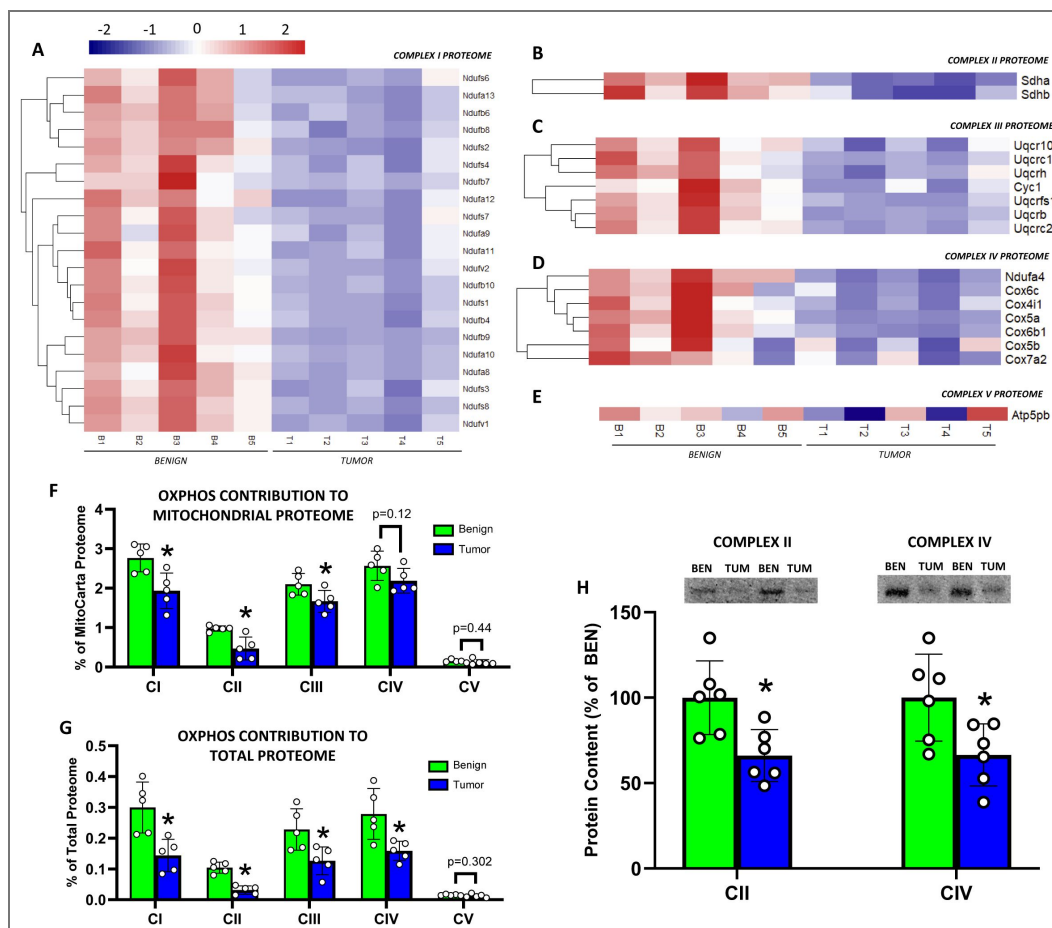
Supplementary Figure 2. Unique HER2-driven tumor and benign mammary tissue proteins.

A) Venn diagram depicting 741 and 600 unique proteins identified in benign mammary tissue and HER2-driven tumors respectively, **B)** KEGG analysis of all detected unique benign mammary tissue proteins, **C)** Top 10 pathways (Reactome 2022), **D)** Top 10 gene ontologies (GO Biological Process) in benign mammary tissue, **E)** KEGG analysis of unique HER2-driven tumor proteins, **F)** Top 10 pathways (Reactome 2022), **G)** Top 10 gene ontologies (GO Biological Process) in HER2-driven tumors. KEGG, pathway and GO analyses were conducted on all mammary and tumor proteins detected by quantitative label-free proteomics. The Enrichr open access analysis web tool was used to generate panels **C,D,F** and **G** which were sorted by p-value.



Supplementary Figure 3. Mitochondrial enrichment in benign mammary tissue and HER2-driven mammary tumors.

A) Total and mitochondrial (Mouse MitoCarta 3.0) absolute protein abundance assessed by quantitative label-free proteomics. **B)** Mitochondrial enrichment factor (MEF) calculated as the ratio of all detected MitoCarta proteins relative to the underlying total proteome per sample (n=5/group). Statistical analysis was conducted using an unpaired two tailed Student's t-test (*p<0.05). In panel **A**, *# denotes significant differences between both total and MitoCarta proteomes between groups.

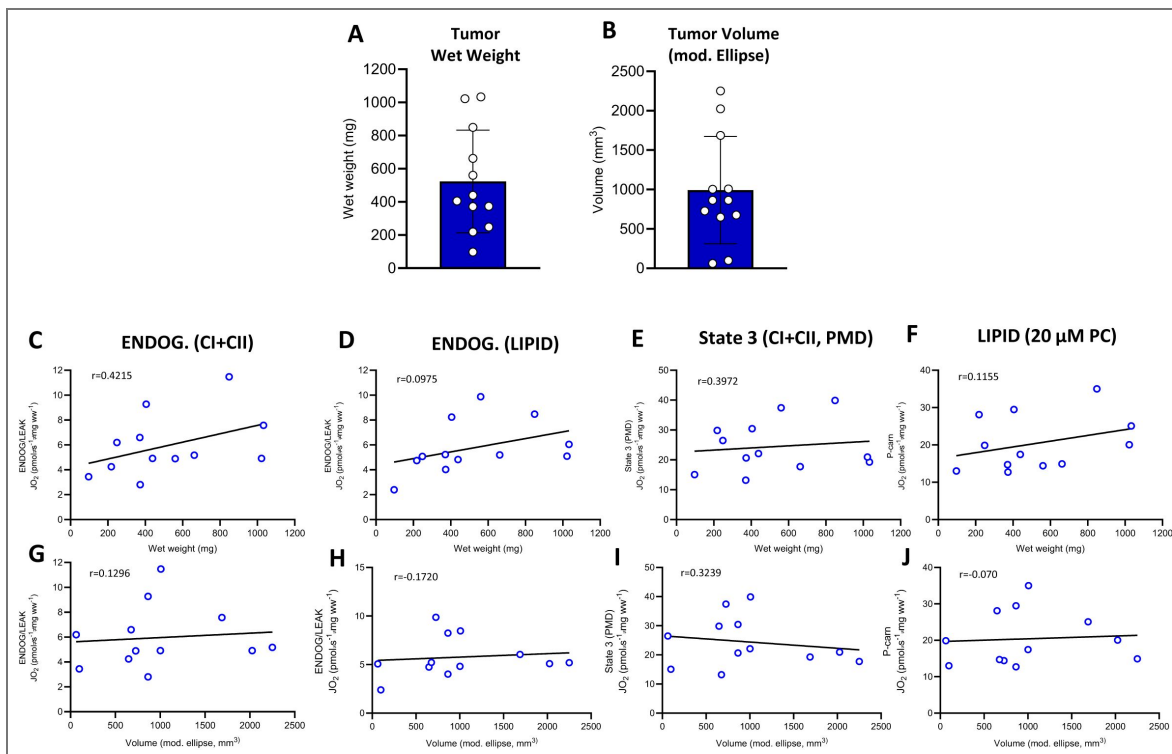


Supplementary Figure 4. Proteomic analysis of detected OXPHOS subunits.

Heatmaps showing subunits detected for **A**) Complex I, **B**) Complex II, **C**) Complex III, **D**) Complex IV, **E**) Complex V (ATP synthase) using quantified label-free proteomics. **F**) OXPHOS Complex I-V absolute abundance as a percentage of the mitochondrial proteome (n=5/group). **G**) OXPHOS complex I-V absolute abundance as a percentage of the total detected proteome (n=5/group). **H**) Representative and quantified western blots for Complex II and Complex IV expressed as percentage of benign. Data in panels **F,G,H** are presented as means $\pm$ SD and analyzed with an unpaired, two tailed Student's t-test (*p<0.05).

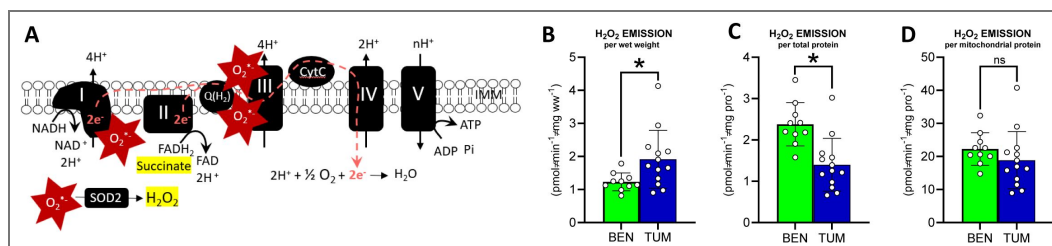
Supplementary Figure 5. Correlation of tumor size and volume with maximal carbohydrate and lipid-supported respiration rates.

A) Tumor wet weight (whole tumor post excision), **B)** Tumor volume calculated as a modified ellipse ($V = 1/2(\text{length} \times \text{width}^2)$). **C-F)** Endogenous (no substrate, CI+CII protocol), Endogenous (no substrate, lipid protocol), State 3 (5 mM pyruvate, 2 mM malate, 5 mM ADP) and maximal lipid-supported (2 mM malate, 5 mM ADP, 20 μ M palmitoyl-carnitine) respiration rates correlated with tissue wet weight. **G-J)** Same respiration data as **(C-F)** but correlated with tumor volume. The Pearson correlation coefficient (R) values are listed for each correlation.



Supplementary Figure 6. Reactive oxygen species emission in benign mammary tissue and HER2-driven mammary tumors.

A) Schematic depicting major sites of superoxide ($\text{O}_2^{\cdot -}$) emission from respiratory complexes I and III. Superoxide dismutase 2 (SOD2) subsequently catalyzes the dismutation of $\text{O}_2^{\cdot -}$ into hydrogen peroxide (H_2O_2) which can be measured fluorometrically. **B)** H_2O_2 emission per wet tissue weight, **C)** H_2O_2 emission per total protein, **D)** H_2O_2 emission per mitochondrial protein (MEF, mitochondrial enrichment factor). Statistical analyses were conducted using an unpaired, two-tailed Student's t-test (* $p < 0.05$).



Abbrev.	Protein	Pathway/Function	Fc (vs BEN)	p.adj
PGK1	Phosphoglycerate kinase	Glycolysis	2.48	0.068
KPYM	Pyruvate kinase	Glycolysis	3.17	0.048
MP2K4	Mitogen activated protein kinase kinase 4	MAPK Pathway	3.65	0.012
RL38	Large ribosomal subunit protein eL38	Protein synthesis/translation	4.11	0.078
PGTA	Geranylgeranyl transferase type-2 subunit alpha	Cysteine prenylation of Rab proteins	4.28	0.067
RI23	Large ribosomal subunit protein uL14	Protein synthesis/translation	4.41	0.042
CADH1	Cadherin-1	Cell adhesion and motility, epithelial cell marker	12.75	0.084
LMCD1	LIM and cysteine rich domains protein 1	Transcriptional cofactor for GATA6 involved in cell proliferation and metastasis (MMP, EGFR)	15.26	0.077
MPRI	Cation independent mannose 6 phosphate receptor	Lysosomal/Golgi trafficking, associates with IGFR2	22.3	0.093

Supplementary Table 1. List of 9 significantly upregulated proteins between tumor and benign samples, sorted in order of fold change from benign (smallest to largest fold change).

General pathways or functions are listed per protein based on ID mapping from the Uniprot protein sequence database. Adjusted p-values are listed per target.

Data availability

Raw RNA sequencing and proteomics data were deposited in GEO (accession number: GSE315552) and PRIDE databases (accession number: PXD071897) respectively.

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Peer reviews

Reviewer #1 (Public review):

Summary:

In this manuscript, Frangos et al. used a transcriptomic and proteomic approach to characterise changes in HER2-driven mammary tumours compared to healthy mammary tissue in mice. They observed that mitochondrial genes, including OXPHOS regulators, were among the most down-regulated genes and proteins in their datasets. Surprisingly, these were associated with higher mitochondrial respiration, in response to a variety of carbon sources. In addition, there seems to be a reduction in mitochondrial fusion and an increase in fission in tumour tissues compared to healthy tissues.

Strengths:

The data are clearly presented and described.

The author reported very similar trends in proteomic and transcriptomic data. Such approaches are essential to have a better understanding of the changes in cancer cell metabolism associated with tumorigenesis.

The authors provided a direct link between HER2 inhibition and OXPHOS, strengthening the mechanistic aspect of the work.

Weaknesses:

The manuscript would have benefited from more ex-vivo approaches to further dissect mechanistic links and resolve the contradiction of elevated respiration with reduced expression of most associated proteins (but these points are clearly articulated in the discussion).

The results presented support the authors' conclusions, and limitations are addressed in the discussion. This work will likely impact the progression of the field, and the provided data will benefit the scientific community.

Comments on revisions:

The authors addressed all my concerns.

<https://doi.org/10.7554/eLife.104079.2.sa2>

Reviewer #2 (Public review):

Frangos et al present a set of studies aiming to determine mechanisms underlying initiation and tumour progression. Overall, this work provides some useful datasets, further establishing mitochondrial dysfunction during the cellular transformation process.

A key strength is the coordinated analysis of transcriptomics and proteomics from tumour samples derived from a Neu-dependent mouse model for breast cancer. This analysis provides rigorous datasets that show robust patterns, including down-regulation across many components of mitochondrial OXPHOS that were generally consistent at both the mRNA and protein level. Parallel analysis of corresponding tumour samples thereby clearly shows the opposite trend of increased mitochondrial function, which is unexpected. As such, this work further establishes altered mitochondrial phenotypes in tumour contexts and further illustrates that mitochondrial function is not necessarily always tightly correlated with mitochondrial gene expression patterns.

Several key weaknesses remain. It remains unclear how increased mitochondrial function is being sustained despite wide decreases in mRNA and protein levels of OXPHOS components. In terms of mechanism, the study confirmed that pharmacologic EGFR inhibition decreases OXPHOS in a EGFR-dependent breast cancer line. However, it remains unclear if the cell culture system recapitulates other key observations of the tumour model (namely decreased expression with increased function).

Therefore, the mechanistic basis of increased mitochondrial function in light of decreased mitochondrial content remains speculative, as does the role of these changes for tumour initiation or progression.

Comments on revisions:

We agree with the overall findings of the study and appreciate that the claims in text and title have been appropriately toned down.

As additional suggestions eg for presentation, many of the graphics/labels are still too small to be useful. It would be interesting to see if this cell line is similar to the tumours in terms of all the phenotypes. The lapatinib experiment was good. I wonder how quick this drug affects the mitochondria. Also it would be interesting to see if these cells have higher OXPHOS than other non-transformed breast epithelial cells.

The WB on oxphos components is good with ab110413 but this looks like many subunits are detected so this should be made clear.

<https://doi.org/10.7554/eLife.104079.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this manuscript, Frangos et al. used a transcriptomic and proteomic approach to characterise changes in HER2-driven mammary tumours compared to healthy mammary tissue in mice. They observed that mitochondrial genes, including OXPHOS regulators, were among the most down-regulated genes and proteins in their datasets. Surprisingly, these were associated with higher mitochondrial respiration, in response to a variety of carbon sources. In addition, there seems to be a reduction in mitochondrial fusion and an increase in fission in tumours compared to healthy tissues.

Strengths:

The data are clearly presented and described.

The author reported very similar trends in proteomic and transcriptomic data. Such approaches are essential to have a better understanding of the changes in cancer cell metabolism associated with tumorigenesis.

Weaknesses:

(1) This study, despite being a useful resource (assuming all the data will be publicly available and not only upon request) is mainly descriptive and correlative and lacks mechanistic links.

We appreciate this point. While the primary goal of our study was to assess mitochondrial adaptations with HER2-driven tumorigenesis, we agree strengthening the mechanistic interpretation would improve the impact of the data. To address this, we have provided experiments demonstrating HER2 inhibition in NF639 cells with lapatinib suppresses respiratory capacity, directly supporting the interpretation that HER2 activity regulates respiratory function (Figure 10). We have expanded the discussion appropriately (lines 378-394). Both raw RNA-seq and proteomic data were deposited through GEO and the PRIDE repositories (accession numbers included in Data Availability Statement).

(2) It would be important to determine the cellular composition of the tumour and healthy tissue used. Do the changes described here apply to cancer cells only or do other cell types contribute to this?

We thank the reviewer for this suggestion; we have added experiments that have directly addressed this concern.

Cell type composition analysis by immunofluorescence was added (Figure 6) where we quantified epithelial, mesenchymal, endothelial, immune and stromal populations in our benign mammary tissue and tumor samples. We found no major shift in the dominant cell types that would confound transcriptomic data in whole tissues.

We integrated immunofluorescence data with a publicly available scRNA-seq dataset from human breast tumors which allowed us to estimate cell-type-specific expression of OXPHOS genes in our own samples. Despite the possibility of species differences, this is the only dataset of its kind, and we used this to generate an estimate of cell type weighted OXPHOS mRNA expression (Figure 6). This revealed that epithelial cells are likely the dominant contributors to OXPHOS gene expression for CI+II. All calculations are delineated in the Methods section.

(3) Are the changes in metabolic gene expression a consequence of HER2 signalling activation? Ex-vivo experiments could be performed to perturb this pathway and determine cause-effects.

Thank you for this suggestion – we have included an experiment directly testing this concept. We assessed mitochondrial respiration in NF639 HER2-driven mammary tumor epithelial cells in the presence or absence of the well-described dual tyrosine kinase inhibitor lapatinib. Lapatinib reduced basal, CI-linked and CI+II linked respiration without compromising mitochondrial integrity or coupling, demonstrating that HER2 activation regulates respiration in our model. This data is presented in Figure 10, and a new section has been added to the discussion describing the implications of this finding in the context of the current literature (lines 378-394).

(4) The data of fission/fusion seem quite preliminary and the gene/protein expression changes are not so clear cut to be a convincing explanation that this is the main reason for the increased mitochondria respiration in tumours.

We agree mitochondrial morphology and dynamics alone cannot fully account for the observed respiratory phenotype – this was emphasized in the discussion but has since been further clarified (lines 365-377). We retained the TEM and dynamics gene/protein data because they do support morphological differences consistent with enhanced fission. However, we have revised the tone of our interpretation to more explicitly acknowledge that these findings are correlative, and the updated discussion now emphasizes that the increased respiratory capacity in tumors is likely driven by multiple converging mechanisms.

Reviewer #2 (Public review):

Frangos et al present a set of studies aiming to determine mechanisms underlying initiation and tumour progression. Overall, this work provides some useful insights into the involvement of mitochondrial dysfunction during the cellular transformation process. This body of work could be improved in several possible directions to establish more mechanistic connections.

(5) The interesting point of the paper: the contrast between suppressed ETC components and activated OXPHOS function is perplexing and should be resolved. It is still unclear if activated mitochondrial function triggers gene down-regulation vs compensatory functional changes (as the title suggests). Have the authors considered reversing the HER2-derived signals e.g. with PI3K-AKT-MTOR or ERK inhibitors to potentially separate the expression vs. functional phenotypes? The root of the OXPHOS component down-regulation should also be traced further, e.g. by probing into levels of core mitochondrial biogenesis factors. Are transcript levels of factors encoded by mtDNA also decreased?

We appreciate this insight and agree that the discordance between mitochondrial content and function is fascinating and have addressed the concerns above in the following manner:

- We have altered the title – we agree we cannot definitively say that the enhanced respiratory capacity observed is compensatory.
- We have added experiments in NF639 cells in the presence of lapatinib, a tyrosine kinase inhibitor to interrogate whether HER2 is necessary for our functional outcome of interest – the enhanced respiratory capacity in the tumors. Lapatinib significantly suppressed respiration (Figure 10) demonstrating HER2 signaling directly regulates mitochondrial respiration.
- We have expanded the discussion to provide further comment on potential explanations for increased respiratory function and low mitochondrial content.

(6) The second interesting aspect of this study is the implication of mitochondrial activation in tumours, despite the downregulation of expression signatures, suggestive of a positive role for mitochondria in this tumour model. To address if this is correlative or causal, have the authors considered testing an OXPHOS inhibitor for suppression of tumorigenesis?

Previous studies have eloquently highlighted that directly or indirectly inhibiting mitochondria can suppress growth in HER2-driven breast cancer (PMID:31690671) or alternatively, amplification of mt-HER2 enhances tumorigenesis (PMID: 38291340). In many solid tumors, this is the concept of preclinical and clinical studies using IACS-010759 or similar inhibitors of OXPHOS which do suppress growth but have significant off target effects in healthy tissues (PMID: 36658425, 3580228). We have expanded the discussion to ensure the reader is aware of these previous contributions and highlighted the importance of future work delineating the role of enhanced respiratory function in HER2-driven mammary cancer (lines 378-394).

(7) A number of issues concerning animal/ tumour variability and further pathway dissection could be explored with in vitro approaches. Have the authors considered deriving tumour derived cell cultures, which could enable further confirmations, mechanistic drug studies and additional imaging approaches? Culture systems would allow alternative assessment of mitochondrial function such as Seahorse or flow cytometry (mitochondrial potential and ROS levels).

We thank the reviewer for this suggestion – we have addressed this in part by using the NF639 HER2 driven tumor epithelial line which demonstrated that HER2 regulates our observed respiratory response. Unfortunately, the addition of tumor derived cell cultures was

not feasible or within the scope of our study. Animal and tumor variability has been clarified in the Methods section (lines 424-429). Mitochondrial respiration experiments were performed in paired tissue (benign and tumor from same mouse). Transcriptomic, proteomic and histological analyses were performed on tumors and benign samples from different mice due to tissue limitations.

(8) The study could be greatly improved with further confirmatory studies, eg immunoblotting for mitochondrial components with parallel blots for phospho-signalling in the same samples. It would be interesting if trends could be maintained in tumour-derived cell cultures. It is notable that OXPHOS protein/transcript changes are more consistent (Figure 5, Supplementary Figure 4) than mitochondrial dynamics /mitophagy factors (Figure 8). Core regulatory factors in these pathways should be confirmed by conventional immunoblotting.

We thank the reviewer for this thoughtful comment. While we agree that additional confirmatory studies can be valuable, due to tissue quantity constraints and the number of assays required for our multi-omics analysis, extensive additional blots were not feasible. However, we had sufficient protein to provide select OXPHOS proteins to verify the proteomic data (now provided in S-Fig.4H). Furthermore, we have plotted the fold change of genes and proteins detected in both datasets and added this to Figure 4 (4A, B), further highlighting the consistency between our transcriptomic and proteomic findings. We believe that the highly consistent and concordant nature of our datasets collectively provides strong support for our central objective - determining whether mitochondrial content and respiratory function correlate in HER2-driven mammary tumors. The reproducibility of OXPHOS-related changes reinforces the robustness of our observations. We also appreciate the reviewer's insight that OXPHOS alterations appear particularly consistent. In response, we have edited the discussion to further emphasize this point, especially in relation to the distinctive pattern observed for Complex V, which showed greater preservation relative to Complexes I-IV across several methods (lines 348-364). We comment on how this stoichiometric shift may contribute to intrinsic respiratory activation despite reduced mitochondrial content.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

Further Minor points.

(9) It would be helpful to know further details regarding the source of the tumour samples, particularly for the proteomics (N=5) and transcriptomics (N=6) datasets, since the exact timepoint of tissue harvest and number of tumours/mouse varied, according to the methods section. Were all samples from the omics studies from different mice (ie 11 mice)? B4 and B6 seem like outliers in mitochondrial transcriptomes. Are these directly paired eg with T4 and T6? Are the side-by-side pairs of Ben and Tum samples for blots in Figure 1 and Supplementary Figure 1 from the same mouse.

This has been clarified in the Methods section (lines 424-429). Mitochondrial respiration experiments were performed in paired tissue (benign and tumor from same mouse). Transcriptomic, proteomic and histological analyses were performed on tumors and benign samples from different mice due to tissue limitations.

(10) Further references and details are needed to support the methodology of the mitochondrial function tests (eg. nutrients vs pairing with complexes). What was the time point of nutrient supplementation? It would seem that the lipid substrates should take longer to activate OXPHOS than pyruvate/malate or succinate. Is this the case? Is there speculation as to why succinate supplementation is much more active than

pyruvate+malate? What is +MD in Figure 6? The rationale for pooling data for Figure 7A is unclear since the categories appear to overlap: (pyruvate, malate, ADP) vs. (palmitoyl-carnitine, malate, ADP).

Thank you for this comment. We have expanded the methods (lines 515-531) to provide additional detail on the mitochondrial respiration protocol. Briefly, permeabilized tissues were exposed to substrates delivered at supraphysiological concentrations in a sequential protocol lasting ~30–60 minutes. Under these conditions, mitochondrial respiration reflects the maximal capacity to utilize each substrate rather than the physiological time course of substrate mobilization or uptake that would occur in vivo with the influence of blood flow and transport/substrate availability limitations.

(11) Many of the figures were blurry (Figure 1F, 2B) or had labels that were too small to be effective (Figures 1G, H, 2D-G, 3E-G, 5E-I, 7C, 8B).

The font size of figure labels has been increased where possible and all figures have been exported to maximize resolution.

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